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SOC2040 SP QUIZ 5 & 6 COMBINED SPRING 2024

Started on	2024-05-05 17:42
State	Finished
Completed on	2024-05-19 15:48
Time taken	13 days 22 hours
Marks	0.00/3008.00
Grade	**0.00** out of 20.00 (**0**%)

Question **1**
Not answered
Marked out of 20.00

Given the following program, Draw the process graph and Write the outputs of both parent and child processes.

```
int main()
  {
      fork();
      fork();
      fork();
      printf(“hello\n”0;
      exit(0);
  }
```

Select one:

1. hello
hello
hello
hello
hello
hello
hello

2. hello
hello
hello

3. hello
hello
hello
hello

4. hello
hello
hello
hello
hello
hello
hello
hello

The correct answer is: hello
hello
hello
hello
hello
hello
hello
hello

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171	172	173	174	175
176	177	178	179	180
181	182	183	184	185
186	187	188	189	190
191	192	193	194	195
196	197	198	199	200
201	202	203	204	205
206	207	208	209	210
211	212	213	214	215
216	217	218	219	220
221	222	223	224	225
226	227	228	229	230
231	232	233	234	235
236	237	238	239	240
241	242	243	244	245
246	247	248	249	250
251	252	253	254	255
256	257	258	259	260
261	262	263	264	265
266	267	268	269	270
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Show one page at a time

Finish review

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Course Info

Syllabus

Participants list

Grade/Attendance

Offline-Attendance

Grades

Question 2
Not answered
Marked out of 10.00

Consider a virtual memory system having a 64-bit virtual address space partitioned into 8KB pages.

Determine the number of bits in VPN and VPO

Select one:

1. VPN=56 bits   VPO=8 bits
2. VPN=51 bits   VPO=13 bits
3. VPN=32 bits   VPO=32 bits
4. VPN=13 bits   VPO=51 bits

The correct answer is: VPN=51 bits   VPO=13 bits

Question 3
Not answered
Marked out of 5.00

.... protects the address space of each process from corruption by other processes.

Select one:

1. Virtual Memory
2. Direct Mapped Cache
3. Fully Associative Cache
4. Set Associative Cache

The correct answer is: Virtual Memory

Question 4
Not answered
Marked out of 10.00

Given a Virtual Memory System having 32-bit virtual address, 4KB Page size and using a single Page table, determine the number of bits in VPN and VPO fields of the virtual address.

Select one:

1. VPN=24 bits VPO= 8 bits
2. VPN=12 bits VPO= 20 bits
3. VPN=20 bits VPO= 12 bits
4. VPN=16 bits VPO= 16 bits

The correct answer is: VPN=20 bits VPO= 12 bits

Question 5
Not answered
Marked out of 5.00

Identify Class of thread-unsafe functions: ....

Select one:

1. Functions that do not keep state across multiple invocations
2. Functions that protect shared variables
3. Functions that return a pointer to a static variable
4. Functions that do not call thread-unsafe functions

The correct answer is: Functions that return a pointer to a static variable

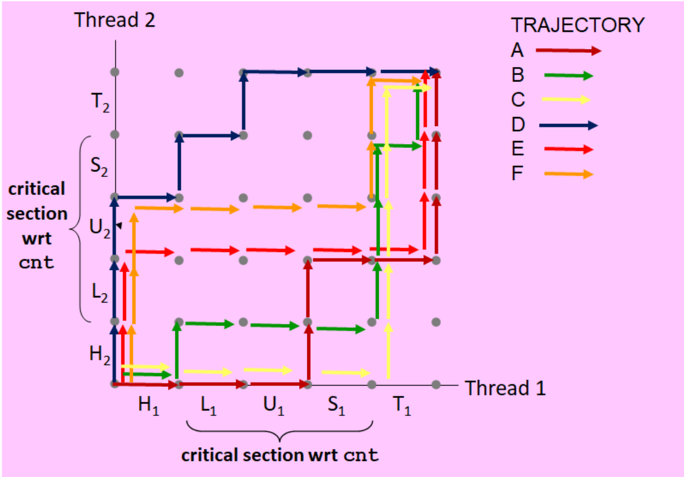
Students Notifications
Others

Activities/Resources

Question 6
Not answered
Marked out of 30.00

For the progress graph given in Figure , classify the following trajectories

A, B, C, D, E, F as safe trajectories.



Select one:
- 1. B, C, D
- 2. A, E, F
- 3. A, B, C
- 4. D, E, F

The correct answer is: B, C, D

Question 7
Not answered
Marked out of 10.00

POSIX function to initialize semaphore mutex variable to 1 for thread synchronization is ....

Select one:
- 1. sem_init(&mutex, 1, 1);
- 2. mutex=1;
- 3. sem_init(&mutex, 0, 1);
- 4. sem_init(&mutex, 1, 0);

The correct answer is: sem_init(&mutex, 0, 1);

Question 8
Not answered
Marked out of 5.00

Example of a synchronous interrupt is ....

Select one:
- 1. interrupt on NMI pin
- 2. keyboard interrupt
- 3. page fault
- 4. timer interrupt

The correct answer is: page fault

Question 9
Not answered
Marked out of 5.00

The POSIX library function to terminate the current thread by a peer thread is ………………………

Select one:
1. pthread_detach
2. pthread_join
3. pthread_cancel
4. pthread_exit

The correct answer is: pthread_cancel

Question 10
Not answered
Marked out of 5.00

The POSIX library function to perform V operation on a semaphore is ………………………

Select one:
1. sem_wait
2. sem_assign
3. sem_init
4. sem_post

The correct answer is: sem_post

Question 11
Not answered
Marked out of 10.00

POSIX stands for ……………………

Answer:

The correct answer is: Portable Operating System Interface

Question 12

Not answered

Marked out of 30.00

Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Show atleast SIX Unsafe Trajectories in the progress graph which lead to improper thread synchronization.

```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

1. H2,L2,U2,S2,T2,H1,L1,U1,S1,T1
H1,H2,L1,U1,S1,L2,U2,S2,T2,T1
H2,L2,U2,S2,H1,L1,U1,S1,T1,T2
H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,U1,S1,L2,U2,S2,T2,T1
H1,L1,U1,S1,T1,H2,L2,U2,S2,T2

2. H2,H1,L2,U2,S2,L1,U1,S1,T1,T2
H2,L2,U2,S2,T2,H1,L1,U1,S1,T1
H1,H2,L1,U1,S1,L2,U2,S2,T2,T1
H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,U1,S1,L2,U2,S2,T2,T1
H2,L2,U2,S2,H1,L1,U1,S1,T1,T2

3. H1,H2,L1,U1,S1,T1,L2,U2,S2,T2
H2,L2,U2,S2,T2,H1,L1,U1,S1,T1
H1,H2,L1,U1,S1,L2,U2,S2,T2,T1
H2,L2,U2,S2,H1,L1,U1,S1,T1,T2
H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,U1,S1,L2,U2,S2,T2,T1

4. H1,L1,H2,L2,U1,S1,T1,U2,S2,T2
H1,H2,L1,U1,L2,S1,U2,S2,T2,T1
H2,L2,U2,H1,L1,U1,S2,S1,T1,T2
H1,H2,L2,L1,U2,S2,U1,S1,T2,T1
H2,H1,L1,L2,U1,U2,S1,S2,T2,T1
H1,L1,H2,L2,U1,U2,S1,S2,T1,T2

The correct answer is: H1,L1,H2,L2,U1,S1,T1,U2,S2,T2
H1,H2,L1,U1,L2,S1,U2,S2,T2,T1
H2,L2,U2,H1,L1,U1,S2,S1,T1,T2

H1,H2,L2,L1,U2,S2,U1,S1,T2,T1
H2,H1,L1,L2,U1,U2,S1,S2,T2,T1
H1,L1,H2,L2,U1,U2,S1,S2,T1,T2

Question 13
Not answered
Marked out of 5.00

.... causes change in control flow in response to a system event (i.e., change in system state).

Answer:

The correct answer is: Exception

Question 14
Not answered
Marked out of 20.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 32 bits wide.

Physical addresses(PA) are 32 bits wide.

The page size is 4KB.

The TLB is four-way set associative with 1024 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 32-byte line(block)size and 4K total blocks.

Cache block is byte addressable.

Determine the number of bits in the following

PPN = ... bits, PPO = ... bits, CO = ... bits, CI = ... bits, CT = ... bits

Select one:
1. PPN=12 bits, PPO=20 bits, CO=5 bits, CI=10 bits, CT=17 bits
2. PPN=20 bits, PPO=12 bits, CO=5 bits, CI=10 bits, CT=16 bits
3. PPN=20 bits, PPO=12 bits, CO=10 bits, CI=5 bits, CT=17 bits
4. PPN=20 bits, PPO=12 bits, CO=5 bits, CI=10 bits, CT=17 bits

The correct answer is: PPN=20 bits, PPO=12 bits, CO=5 bits, CI=10 bits, CT=17 bits

Question 15
Not answered
Marked out of 10.00

Given a Virtual Memory System having 32-bit virtual address, 4KB Page size and using a single Page table.

Determine the number of entries in the page table(i.e., number of PTEs).

Select one:
1. PTEs = 2Mega entries
2. PTEs = 1Mega entries
3. PTEs = 2Giga entries
4. PTEs = 1Giga entries

The correct answer is: PTEs = 1Mega entries

Question 16
Not answered
Marked out of 20.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

Cache block is byte addressable.

Determine the number of bits in the following

PPN = ... bits, PPO = ... bits, CO = ... bits,   CI = ... bits,   CT = ... bits

Select one:

1. PPN=8 bits, PPO=4 bits, CO=4 bits, CI=2 bits, CT=6 bits
2. PPN=6 bits, PPO=6 bits, CO=6 bits, CI=2 bits, CT=4 bits
3. PPN=6 bits, PPO=6 bits, CO=2 bits, CI=4 bits, CT=6 bits
4. PPN=8 bits, PPO=6 bits, CO=6 bits, CI=2 bits, CT=6 bits

The correct answer is: PPN=6 bits, PPO=6 bits, CO=2 bits, CI=4 bits, CT=6 bits

Question 17
Not answered
Marked out of 5.00

The POSIX library function to reap a peer thread by a main thread or another peer thread is ………………………………………….

Select one:

1. pthread_create
2. pthread_detach
3. pthread_self
4. pthread_join

The correct answer is: pthread_join

Question 18
Not answered
Marked out of 10.00

A function is .... iff it will always produce correct results when called repeatedly from multiple concurrent threads

Answer:

The correct answer is: thread-safe

Question 19
Not answered
Marked out of 10.00

Multithreaded caching Web proxy is an example of ....

Select one:
1. Travelling Salesman Problem
2. Producer-Consumer Problem
3. Dining Philosophers Problem
4. Readers-Writers Problem

The correct answer is: Readers-Writers Problem

Question 20
Not answered
Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one infeasible output

```
int main()
{
    printf("L0\n");
    if (fork() == 0) {
        printf("L1\n");
        if (fork() == 0) {
            printf("L2\n");
        }
    }
    printf("Bye\n");
}
```

Select one:

1. L0
L1
Bye
Bye
L2
Bye

2. L0
L1
Bye
L2
Bye
Bye

3. L0
Bye
L1
Bye
Bye
L2

4. L0
Bye
L1
L2
Bye
Bye

The correct answer is: L0
Bye
L1
Bye
Bye
L2

Question 21
Not answered
Marked out of 10.00

A .... is a useful tool for capturing the partial ordering of statements in a concurrent program.

Answer: ✗

The correct answer is: process graph

Question 22
Not answered
Marked out of 10.00

Consider a virtual memory system having a 32-bit virtual address space partitioned into 16KB pages.

Determine the number of bits in VPN and VPO

Select one:

- 1. VPN=14 bits VPO=18 bits
- 2. VPN=18 bits VPO=14 bits
- 3. VPN=24 bits VPO=8 bits
- 4. VPN=16 bits VPO=16 bits

The correct answer is: VPN=18 bits VPO=14 bits

Question 23
Not answered
Marked out of 10.00

Linux process hierarchy can be viewed by using .... command

Answer: ✗

The correct answer is: pstree

Question 24

Not answered

Marked out of 30.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

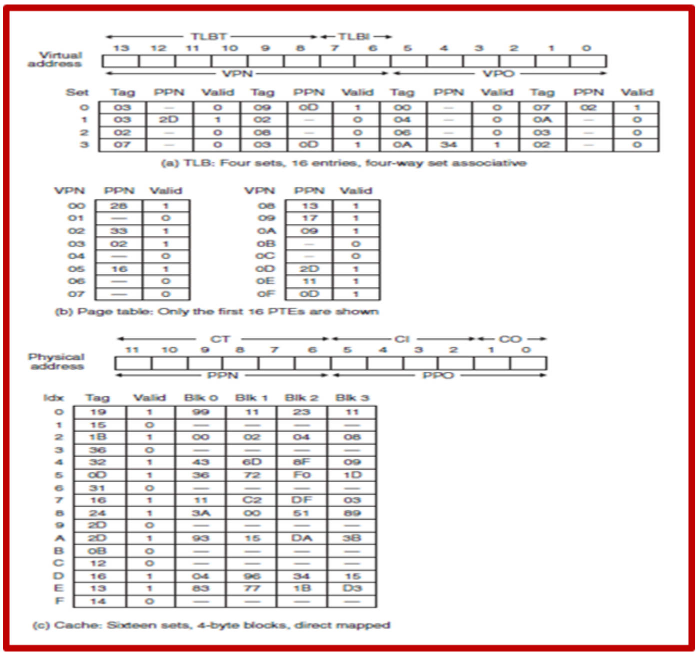
Cache block is byte addressable.

Processor generates a virtual address VA = 0x0320, Translate this to physical address PA using TLB or Page Table

Determine the values (in hex) in the following fields:

VPN ___ VPO ___ TLBI ___ TLBT ____ TLB Hit? __ (Y/N) Page Fault? __(Y/N) PPN: __

Page Table access Required PTA: ______ (Y/N)



Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	—	0	09	0D	1	00	—	0	07	02	1
1	03	2D	1	02	—	0	04	—	0	0A	—	0
2	02	—	0	08	—	0	06	—	0	03	—	0
3	07	—	0	03	0D	1	0A	34	1	02	—	0

(a) TLB: Four sets, 16 entries, four-way set associative

VPN	PPN	Valid	VPN	PPN	Valid
00	28	1	08	13	1
01	—	0	09	17	1
02	33	1	0A	09	1
03	02	1	0B	—	0
04	—	0	0C	—	0
05	16	1	0D	2D	1
06	—	0	0E	11	1
07	—	0	0F	0D	1

(b) Page table: Only the first 16 PTEs are shown

Idx	Tag	Valid	Blk 0	Blk 1	Blk 2	Blk 3
0	19	1	99	11	23	11
1	15	0	—	—	—	—
2	1B	1	00	02	04	08
3	36	0	—	—	—	—
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	—	—	—	—
7	16	1	11	C2	DF	03
8	24	1	3A	00	51	89
9	2D	0	—	—	—	—
A	2D	1	93	15	DA	3B
B	0B	0	—	—	—	—
C	12	0	—	—	—	—
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	—	—	—	—

(c) Cache: Sixteen sets, 4-byte blocks, direct mapped

Select one:

1. VPN = 0x0C, VPO= 0x20, TLBI= 0x00, TLBT= 0x03, TLB Hit? =N, Page Fault?= Y, PPN = ..., PTAR: N
2. VPN = 0x0C, VPO= 0x20, TLBI= 0x03, TLBT= 0x00, TLB Hit? =Y, Page Fault?= Y, PPN = ..., PTAR: N
3. VPN = 0x20, VPO= 0x0C, TLBI= 0x00, TLBT= 0x03, TLB Hit? =N, Page Fault?= N, PPN = ..., PTAR: N
4. VPN = 0x0C, VPO= 0x20, TLBI= 0x03, TLBT= 0x00, TLB Hit? =N, Page Fault?= Y, PPN = ..., PTAR: N

The correct answer is: VPN = 0x0C, VPO= 0x20, TLBI= 0x00, TLBT= 0x03, TLB Hit? =N, Page Fault?= Y, PPN = ..., PTAR: N

Question 25
Not answered
Marked out of 5.00

Identify the relative pathname from the following:

Select one:
1. /usr/bin/vim
2. ../../home/droh/hello.c
3. /usr/include/sys/unistd.h
4. /dev//tty1

The correct answer is: ../../home/droh/hello.c

Question 26
Not answered
Marked out of 5.00

.... = Program context + Kernel context

Answer:

The correct answer is: Process Context

Question 27
Not answered
Marked out of 20.00

Given a Virtual Memory System having 32-bit virtual address, 4KB Page size and using a single Page table with 8 byte PTE (size of each Page table entry), determine the amount of Main Memory required to store a single Page Table.

Select one:
1. 4GB
2. 4MB
3. 8MB
4. 2MB

The correct answer is: 8MB

Question 28
Not answered
Marked out of 10.00

.... function is used by processes to send signals to other processes (including themselves).

Answer:

The correct answer is: kill

Question 29

Not answered

Marked out of 10.00

Consider a virtual memory system having a 64-bit virtual address space partitioned into 16KB pages.

Determine the number of bits in VPN and VPO

Select one:

- 1. VPN=50 bits   VPO=14 bits
- 2. VPN=56 bits   VPO=8 bits
- 3. VPN=32 bits   VPO=32 bits
- 4. VPN=14 bits   VPO=50 bits

The correct answer is: VPN=50 bits   VPO=14 bits

Question 30

Not answered

Marked out of 50.00

Consider the following program where main thread creates 5 threads.

Provide the results provided by the main thread & peer threads.

```
char **ptr1;
char **ptr2;
int  count;

int main()
{
   long i;
   pthread_t tid[5];

   char *msgs1[2] = {
      "Hello from thread0",
      "Hello from thread1"
   };
  char *msgs2[3] = {
      "Hello from thread3",
      "Hello from thread4",
       "Hello from thread5"
   };
   count=100;
   ptr1 = msgs1;
   ptr2 = msgs2;

   for (i = 0; i < 2; i++)
      pthread_create(&tid[i],  NULL, thread1, (void *)i);
  for (i = 2; i < 5; i++)
      pthread_create(&tid[i],  NULL, thread2, (void *)i);
   printf("THREAD MAIN EXITING\n");
   Pthread_exit(NULL);
}
```

```
void *thread1(void *vargp1)
{
long myid1 = (long)vargp1;
 static int cnt1 = 0;

count *= ++cnt1;
printf("[%ld]:  %s (cnt=%d) – count=%d \n",
        myid1, ptr1[myid1], cnt1, count);
 printf("THREAD %ld RETURNING \n", myid1);
 return NULL;
}

void *thread2(void *vargp2)
{
long myid2 = (long)vargp2;
static int cnt2 = 3;

count *= cnt2--;
printf("[%ld]:  %s (cnt=%d) – count=%d \n",  myid2,
ptr2[myid2-2], cnt2, count);
printf("THREAD %ld RETURNING \n", myid2);
return NULL;
}
```

Select one:

1. [0] Hello from thread0(cnt=1)-count=100
   THREAD 0 RETURNING
   [1] Hello from thread1(cnt=2)-count=200
   THREAD 1 RETURNING
   [2] Hello from thread3(cnt=2)-count=600
   THREAD 2 RETURNING
   [3] Hello from thread4(cnt=1)-count=1200
   THREAD 3 RETURNING
   [4] Hello from thread5(cnt=0)-count=1200
   THREAD 4 RETURNING

2. [0] Hello from thread0(cnt=1)-count=101
   THREAD 0 RETURNING
   [1] Hello from thread1(cnt=2)-count=102
   THREAD 1 RETURNING
   [2] Hello from thread2(cnt=3)-count=105
   THREAD 2 RETURNING
   [3] Hello from thread3(cnt=2)-count=107
   THREAD 3 RETURNING
   [4] Hello from thread4(cnt=1)-count=108
   THREAD 4 RETURNING

3. [0] Hello from thread0(cnt=1)-count=100
   THREAD 0 RETURNING
   [1] Hello from thread1(cnt=2)-count=200
   THREAD 1 RETURNING
   [2] Hello from thread3(cnt=2)-count=600
   THREAD 2 RETURNING
   [3] Hello from thread4(cnt=1)-count=1200
   THREAD 3 RETURNING
   [4] Hello from thread5(cnt=0)-count=0
   THREAD 4 RETURNING

4. [0] Hello from thread0(cnt=1)-count=100
   THREAD 0 RETURNING
   [1] Hello from thread1(cnt=2)-count=200
   THREAD 1 RETURNING
   [2] Hello from thread2(cnt=2)-count=600
   THREAD 2 RETURNING
   [3] Hello from thread3(cnt=1)-count=1200
   THREAD 3 RETURNING
   [4] Hello from thread4(cnt=0)-count=1200
   THREAD 4 RETURNING

The correct answer is: [0] Hello from thread0(cnt=1)-count=100
THREAD 0 RETURNING
[1] Hello from thread1(cnt=2)-count=200

```
THREAD 1 RETURNING
[2] Hello from thread3(cnt=2)-count=600
THREAD 2 RETURNING
[3] Hello from thread4(cnt=1)-count=1200
THREAD 3 RETURNING
[4] Hello from thread5(cnt=0)-count=1200
THREAD 4 RETURNING
```

Question 31

Not answered

Marked out of 25.00

Given the following program using a signal handler.

Provide the outputs produced by this program in sequence.

```
int main()
 {
      signal(SIGALRM, handler);

         alarm(100);
         printf("STUDY HARDER........\n");
          while (1) {
           ;

          }
         exit(0);
 }

void handler(int sig)
 {
     static int study = 0;
     printf("STUDY STILL HARDER........\n");
      if (++ study < 4)
              alarm(900);
       else
       {
              printf("RESULT : GRADE A+   WOW!!!!!\n");
              exit(0);
        }
 }
```

Select one:

1. STUDY HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   RESULT : GRADE A+ WOW!!!!!

2. STUDY HARDER……..
   STUDY STILL HARDER……..
   STUDY HARDER……..
   STUDY STILL HARDER……..
   STUDY HARDER……..
   STUDY STILL HARDER……..
   STUDY HARDER……..
   STUDY STILL HARDER……..
   RESULT : GRADE A+ WOW!!!!!

3. STUDY HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY HARDER……..
   RESULT : GRADE A+ WOW!!!!!

4. STUDY HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   STUDY STILL HARDER……..
   RESULT : GRADE A+ WOW!!!!!

The correct answer is: STUDY HARDER……..
STUDY STILL HARDER……..
STUDY STILL HARDER……..

STUDY STILL HARDER……..
STUDY STILL HARDER……..
RESULT : GRADE A+ WOW!!!!!

Question 32
Not answered
Marked out of 5.00

The waiting time required for first bit of the target sector to pass under the read/write head after positioning the head over the required track is called …………………………………….

Select one:
1. rotational latency
2. access time
3. transfer time
4. seek time

The correct answer is: rotational latency

Question 33
Not answered
Marked out of 5.00

………… exception always terminates the current program.

Select one:
1. Trap
2. Abort
3. Fault
4. Any

The correct answer is: Abort

Question 34
Not answered
Marked out of 5.00

…………………………………occurs when there is enough aggregate free memory to satisfy an allocate request, but no single free block is large enough to handle the request.

Select one:
1. External Fragmentation
2. Internal Fragmentation
3. Compaction
4. False fragmentation

The correct answer is: False fragmentation

Question 35
Not answered
Marked out of 5.00

End of line (EOL) indicator used in Windows regular file is ………………………..

Select one:
- 1. Carriage Return (CR)
- 2. Carriage Return followed by Line Feed
- 3. Line Feed(LF)
- 4. Line Feed followed by Carriage Return

The correct answer is: Carriage Return followed by Line Feed

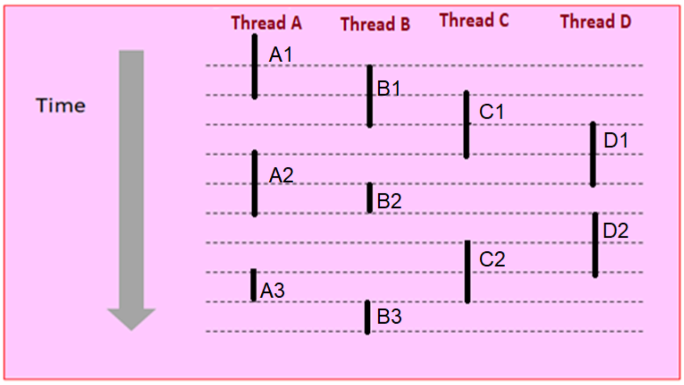
Question 36
Not answered
Marked out of 30.00

Given the following timing diagram showing the execution flows for four different threads A, B, C & D.

You are given two cores on the processor to execute these four threads.

Each core can execute only one thread at a time.

Please provide clearly how these threads will be executed on the two cores of the processor chip.



Select one:
- 1. Core 0 : Thread A1, A2, A3, Thread B1, B2, B3

  Core 1: Thread C1, C2 Thread D1, D2
- 2. Core 0 : Thread A1, A2, A3 Thread C1, Thread D2
  Core 1: Thread B1, B2, B3 Thread D1 Thread C2
- 3. Core 0 : Thread A1, A2, A3 Thread C1, C2
  Core 1: Thread B1, B2, B3 Thread D1, D2
- 4. Core 0 : Thread A1, A3, Thread B2, Thread C1, Thread C2
  Core 1: Thread B1, B3, Thread A2, Thread D1 Thread D2

The correct answer is: Core 0 : Thread A1, A2, A3 Thread C1, Thread D2
Core 1: Thread B1, B2, B3 Thread D1 Thread C2

Question 37
Not answered
Marked out of 10.00

In ...., Logical flows are modelled as state machines that the main program explicitly transitions from state to state as a result of data arriving on file descriptors. Since the program is a single process, all flows share the same address space.

Answer:

The correct answer is: I/O MULTIPLEXING

Question 38
Not answered
Marked out of 5.00

The function to get the PID of a calling process is ....

Select one:
1. getppid
2. getpgrp
3. getpid
4. getppgrp

The correct answer is: getpid

Question 39
Not answered
Marked out of 20.00

Given a Virtual Memory System having 64-bit virtual address, 4KB Page size and using a single Page table with 8 byte PTE (size of each Page table entry), determine the amount of Main Memory required to store a single Page Table.

Select one:
1. 32PB
2. 16GB
3. 32GB
4. 16PB

The correct answer is: 32PB

Question 40
Not answered
Marked out of 10.00

Consider a virtual memory system having a 64-bit virtual address space partitioned into 4MB pages.

Determine the number of bits in VPN and VPO

Select one:
1. VPN=42 bits VPO=22 bits
2. VPN=56 bits VPO=8 bits
3. VPN=22 bits VPO=42 bits
4. VPN=32 bits VPO=32 bits

The correct answer is: VPN=42 bits VPO=22 bits

Question 41
Not answered
Marked out of 5.00

Control flow passes from one process to another process via ....

Select one:
1. Jump instruction
2. Context switching
3. Call instruction
4. Return instruction

The correct answer is: Context switching

Question 42
Not answered
Marked out of 5.00

fork is called once but returns ....

Select one:
1. thrice
2. twice
3. never
4. once

The correct answer is: twice

Question 43
Not answered
Marked out of 5.00

Identify Class of thread-safe functions: ....

Select one:
1. Functions that return a pointer to a static variable
2. Functions that do not thread-unsafe functions
3. Functions that do not protect shared variables
4. Functions that keep state across multiple invocations

The correct answer is: Functions that do not call thread-unsafe functions

Question 44
Not answered
Marked out of 5.00

The POSIX library function to determine its own thread ID by a calling thread is ....

Select one:
1. pthread_create
2. pthread_join
3. pthread_self
4. pthread_detach

The correct answer is: pthread_self

Question 45
Not answered
Marked out of 5.00

The ID number of x86-64 System call   execve   is ………

Select one:
- 1. 60
- 2. 59
- 3. 62
- 4. 57

The correct answer is: 59

Question 46
Not answered
Marked out of 20.00

Given a Virtual Memory System having 48-bit virtual address, 64KB Page size and using a single Page table with 4 byte PTE (size of each Page table entry).

Determine the amount of Main Memory required to store a single Page Table.

Select one:
- 1. 4GB
- 2. 2GB
- 3. 16GB
- 4. 8GB

The correct answer is: 16GB

Question 47
Not answered
Marked out of 15.00

Given the following program, Draw the process graph and determine the total number of child processes created when this program is executed.

```
int main()
{
    fork();
    fork();
    fork();
    printf("hello\n");
    exit(0);
}
```

Select one:
- 1. 7
- 2. 6
- 3. 3
- 4. 8

The correct answer is: 7

Question 48

Not answered

Marked out of 5.00

The read function returns ..... on encountering EOF.

Select one:

1. 0
2. 1
3. -1
4. number of bytes read

The correct answer is: 0

Question 49

Not answered

Marked out of 25.00

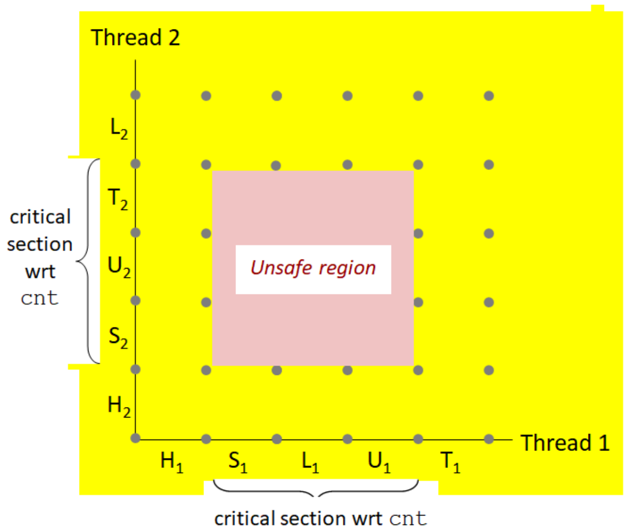
Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Draw the unsafe region in progress graph.

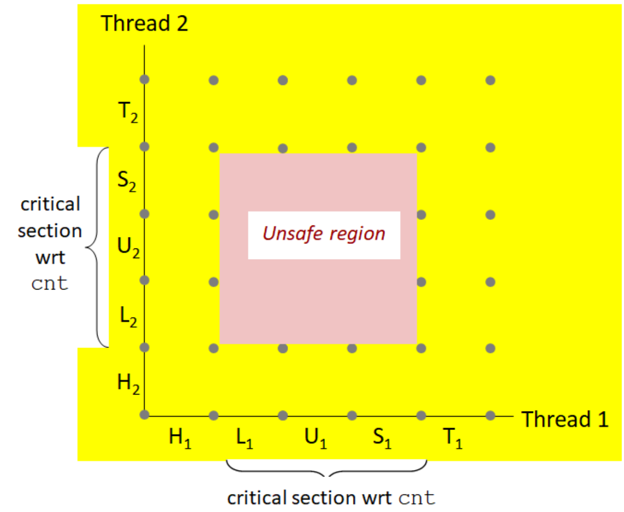
```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =    *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

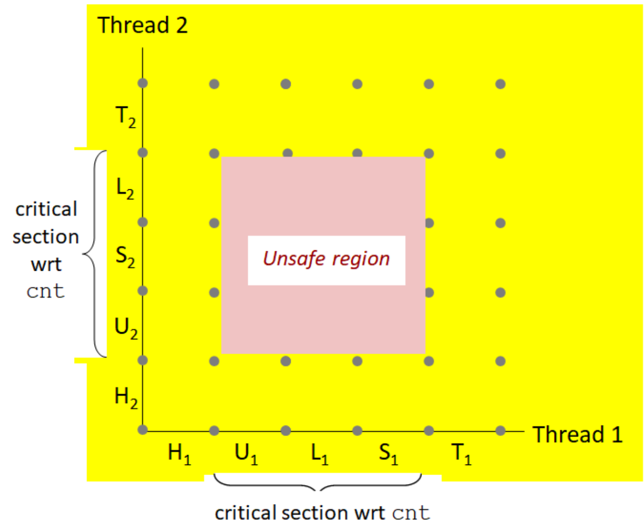
1.



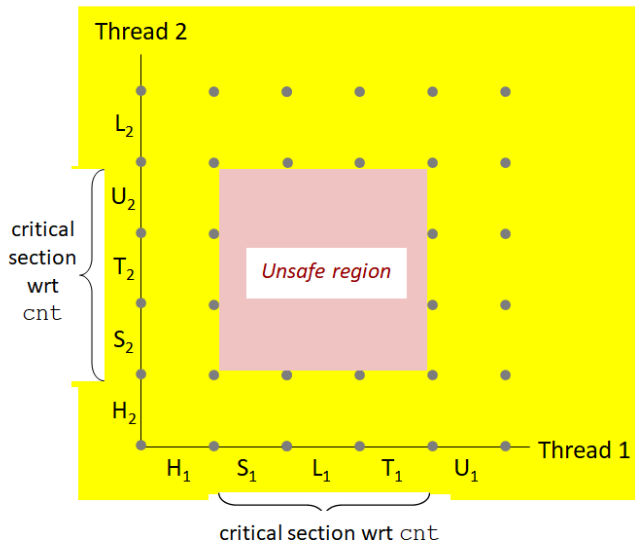
2.



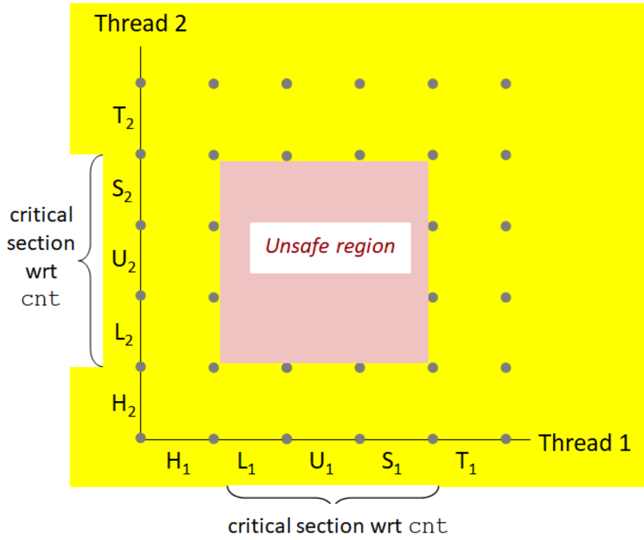
3.



4.



The correct answer is:



Question 50
Not answered
Marked out of 5.00

……………………………………is a small set-associative hardware cache inside the CPU which maps virtual page numbers to physical page numbers

Select one:
1. TLB
2. MMU
3. Cache L2
4. Cache L1

The correct answer is: TLB

Question 51
Not answered
Marked out of 5.00

An example of Synchronous exception Trap is ……………………………………………………………

Select one:
1. Floating point exception
2. Page fault
3. Illegal Instruction
4. Breakpoint

The correct answer is: Breakpoint

Question 52

Not answered

Marked out of 5.00

Synchronous exception Traps are ....

Select one:

1. Unintentional but possibly recoverable
2. Intentional & Unrecoverable
3. Unintentional & Unrecoverable
4. Intentional

The correct answer is: Intentional

Question 53

Not answered

Marked out of 15.00

Given the following program, draw the process graph and Identify atleast one feasible output.

```
int main()
{
 printf("L0\n");
   if (fork() != 0) {
      printf("L1\n");
      if (fork() != 0) {
         printf("L2\n");
            }
   }
   printf("Bye\n");
}
```

Select one:

1. L0, Bye, L1, Bye, Bye, L2
2. L0, Bye, Bye, L1, Bye, L2
3. L0, L1, Bye, L2, Bye, Bye
4. L0, L1, Bye, Bye, Bye, L2

The correct answer is: L0, L1, Bye, L2, Bye, Bye

Question 54

Not answered

Marked out of 5.00

Identify Class of thread-unsafe functions:

Select one:

1. Functions that call thread-unsafe functions
2. Functions that do not keep state across multiple invocations
3. Functions that do not return a pointer to a static variable
4. Functions that protect shared variables

The correct answer is: Functions that call thread-unsafe functions

Question 55

Not answered

Marked out of 30.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

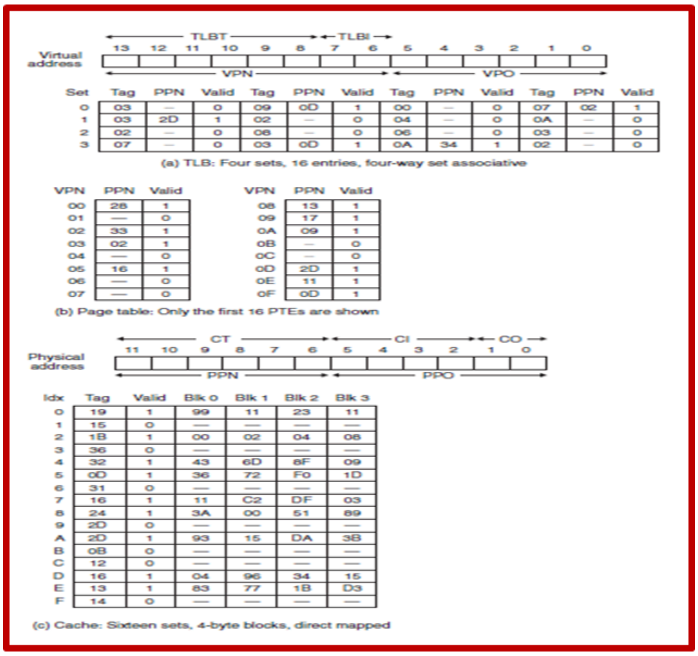
Cache block is byte addressable.

Processor generates a virtual address VA = 0x0020, Translate this to physical address PA using TLB or Page Table

Determine the values (in hex) in the following fields:

VPN ___ VPO ___ TLBI ___ TLBT ____ TLB Hit? __ (Y/N) Page Fault? __(Y/N) PPN: __

Page Table access Required PTA: ______ (Y/N)



Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	—	0	09	0D	1	00	—	0	07	02	1
1	03	2D	1	02	—	0	04	—	0	0A	—	0
2	02	—	0	08	—	0	06	—	0	03	—	0
3	07	—	0	03	0D	1	0A	34	1	02	—	0

(a) TLB: Four sets, 16 entries, four-way set associative

VPN	PPN	Valid	VPN	PPN	Valid
00	28	1	08	13	1
01	—	0	09	17	1
02	33	1	0A	09	1
03	02	1	0B	—	0
04	—	0	0C	—	0
05	16	1	0D	2D	1
06	—	0	0E	11	1
07	—	0	0F	0D	1

(b) Page table: Only the first 16 PTEs are shown

Idx	Tag	Valid	Blk 0	Blk 1	Blk 2	Blk 3
0	19	1	99	11	23	11
1	15	0	—	—	—	—
2	1B	1	00	02	04	08
3	36	0	—	—	—	—
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	—	—	—	—
7	16	1	11	C2	DF	03
8	24	1	3A	00	51	89
9	2D	0	—	—	—	—
A	2D	1	93	15	DA	3B
B	0B	0	—	—	—	—
C	12	0	—	—	—	—
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	—	—	—	—

(c) Cache: Sixteen sets, 4-byte blocks, direct mapped

Select one:

1. VPN = 0x00, VPO= 0x20, TLBI= 0x00, TLBT= 0x00, TLB Hit? =N, Page Fault?= N, PPN = 0x38, PTAR: Y
2. VPN = 0x20, VPO= 0x00, TLBI= 0x00, TLBT= 0x00, TLB Hit? =N, Page Fault?= Y, PPN = 0x28, PTAR: N
3. VPN = 0x00, VPO= 0x20, TLBI= 0x00, TLBT= 0x00, TLB Hit? =N, Page Fault?= N, PPN = 0x28, PTAR: N
4. VPN = 0x20, VPO= 0x00, TLBI= 0x00, TLBT= 0x10, TLB Hit? =N, Page Fault?= Y, PPN = 0x28, PTAR: Y

The correct answer is: VPN = 0x00, VPO= 0x20, TLBI= 0x00, TLBT= 0x00, TLB Hit? =N, Page Fault?= N, PPN = 0x28, PTAR: N

Question 56
Not answered
Marked out of 30.00

Given the following program, Draw the process graph and Identify atleast one infeasible output

```
int main()
{
        pid_t pid;
         printf("COVID 19 CORONA VIRUS ");
         if (fork() == 0 )
          {   printf("BE SAFE & HEALTHY");
                  if (Fork() != 0)
                        printf("PANDEMIC DISEASE");
           }
         if (Fork() !=0)
               printf("MAINTAIN DISTANCING");
         printf("STAY HOME \n");
}
```

Select one:

1. COVID 19 CORONA VIRUS
MAINTAN DISTANCING
BE SAFE & HEALTHY
STAY HOME
STAY HOME
PANDEMIC DISEASE
MAINTAIN DISTANCING
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME

2. COVID 19 CORONA VIRUS
BE SAFE & HEALTHY
MAINTAN DISTANCING
STAY HOME
PANDEMIC DISEASE
STAY HOME
MAINTAIN DISTANCING
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME

3. COVID 19 CORONA VIRUS
BE SAFE & HEALTHY
MAINTAN DISTANCING
STAY HOME
PANDEMIC DISEASE
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME

4. COVID 19 CORONA VIRUS
BE SAFE & HEALTHY
MAINTAN DISTANCING
STAY HOME
STAY HOME
PANDEMIC DISEASE
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME
STAY HOME
MAINTAIN DISTANCING

The correct answer is: COVID 19 CORONA VIRUS
BE SAFE & HEALTHY
MAINTAN DISTANCING
STAY HOME
STAY HOME
PANDEMIC DISEASE
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME
STAY HOME
MAINTAIN DISTANCING

Question 57
Not answered
Marked out of 5.00

The ASCII character corresponding to Carriage Return (CR) is ....

Select one:
- 1. 0x0a
- 2. 0x0b
- 3. 0x0d
- 4. 0x0c

The correct answer is: 0x0d

Question 58
Not answered
Marked out of 20.00

Given a Virtual Memory System having 48-bit virtual address, 4KB Page size and using a single Page table with 8 byte PTE (Page table entry),

determine the amount of Main Memory required to store a single Page Table.

Select one:
- 1. 256GB
- 2. 128GB
- 3. 64TB
- 4. 512GB

The correct answer is: 512GB

Question 59
Not answered
Marked out of 10.00

Identify the correct V function ....

Select one:

1. void V(sem_t *s)
```
{
  sem_incr(&s);
}
```
2. void V(sem_t *s)
```
{
  sem_wait(&s);
}
```
3. void V(sem_t *s)
```
{
  sem_test(&s);
}
```
4. void V(sem_t *s)
```
{
  sem_post(&s);
}
```

The correct answer is: void V(sem_t *s)
```
{
  sem_post(&s);
}
```

Question 60
Not answered
Marked out of 5.00

A ··················is an explicit request made to the kernel to get a particular operating system service via interrupt.

Select one:

1. System Call
2. Exception
3. Signal
4. Interrupt

The correct answer is: System Call

Question 61
Not answered
Marked out of 5.00

································· command is used to display the information about processes currently running, stopped and sleeping.

Select one:

1. ps w
2. ps -la
3. ps -a
4. ps

The correct answer is: ps w

Question 62
Not answered
Marked out of 5.00

A terminated process that has not yet been reaped is called a ....

Answer:

The correct answer is: zombie

Question 63
Not answered
Marked out of 10.00

Calling wait(&status) is equivalent to calling ....

Select one:
1. waitpid(100, &status, 0)
2. waitpid(-1, &status, 0)
3. waitpid(0, &status, 0)
4. waitpid(1, &status, 0)

The correct answer is: waitpid(-1, &status, 0)

Question 64
Not answered
Marked out of 10.00

A .... models the execution of n concurrent threads as a trajectory through an n-dimensional Cartesian space.

Answer:

The correct answer is: progress graph

Question 65
Not answered
Marked out of 5.00

Indicate what happens when the following instruction is executed :

fd = Open("foo.txt", O_WRONLY|O_APPEND, 0);

Select one:
1. Opens the file only for write
2. opens the file for read & append
3. opens the file for both read & write
4. Opens the file only for write & append

The correct answer is: Opens the file only for write & append

Question 66
Not answered
Marked out of 5.00

For each process, the kernel maintains a variable ………………… that points to the top of the heap.

Select one:

1. pc
2. cs
3. sp
4. brk

The correct answer is: brk

Question 67
Not answered
Marked out of 10.00

……… function puts the calling function to sleep until a signal is received by the process.

Answer:

The correct answer is: pause

Question 68
Not answered
Marked out of 5.00

Two processes are termed ………… if their flows overlap in time.

Answer:

The correct answer is: concurrent

Question 69
Not answered
Marked out of 5.00

The system call ……. provides a way to move around in a file without reading or writing any data.

Answer:

The correct answer is: lseek

Question 70
Not answered
Marked out of 20.00

Given a Virtual Memory System having 48-bit virtual address, 4KB Page size and using a single Page table with 8 byte PTE (size of each Page table entry), determine the amount of Main Memory required to store a single Page Table.

Select one:
- 1. 512GB
- 2. 256MB
- 3. 256GB
- 4. 512MB

The correct answer is: 512GB

Question 71
Not answered
Marked out of 5.00

Identify Class of thread-safe functions: ....

Select one:
- 1. Functions that call thread-unsafe functions
- 2. Functions that do not protect shared variables
- 3. Functions that do not keep state across multiple invocations
- 4. Functions that return a pointer to a static variable

The correct answer is: Functions that do not keep state across multiple invocations

Question 72
Not answered
Marked out of 12.00

Reorder the following steps in the correct order to indicate address translation performed by the CPU Hardware when there is a page hit.

a) Cache/memory returns the requested data word to processor

b) MMU fetches PTE from page table in memory

c) Processor sends virtual address to MMU

d) MMU constructs the physical address & sends physical address to cache/memory

Select one:
- 1. c),d),b),a)
- 2. a),b),c),d)
- 3. c),b),d),a)
- 4. b),a),d),a)

The correct answer is: c),b),d),a)

Question 73

Not answered

Marked out of 5.00

Each core can execute a separate process. Scheduling of processes onto cores is done by ………………..

Select one:

1. I/O device
2. Timer
3. application
4. kernel

The correct answer is: kernel

Question 74

Not answered

Marked out of 30.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

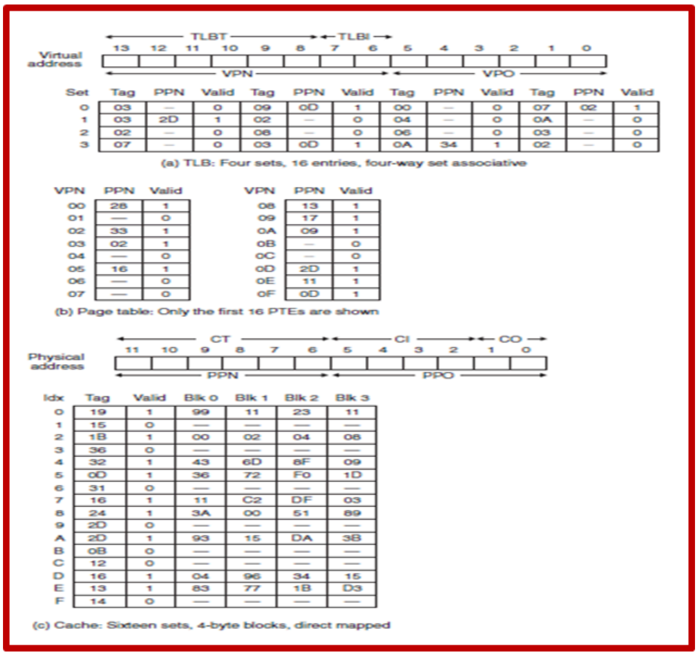
Cache block is byte addressable.

Processor generates a virtual address VA = 0x0020, Translate this to physical address PA using TLB or Page Table

Determine the values (in hex) in the following fields:

Physical Address PA = ..., PPN = ..., PPO= .....

CO=....., CI=......, CT=......, Cache Hit?.... (Y/N), Byte (read from cache)=......



Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	—	0	09	0D	1	00	—	0	07	02	1
1	03	2D	1	02	—	0	04	—	0	0A	—	0
2	02	—	0	08	—	0	06	—	0	03	—	0
3	07	—	0	03	0D	1	0A	34	1	02	—	0

(a) TLB: Four sets, 16 entries, four-way set associative

VPN	PPN	Valid	VPN	PPN	Valid
00	28	1	08	13	1
01	—	0	09	17	1
02	33	1	0A	09	1
03	02	1	0B	—	0
04	—	0	0C	—	0
05	16	1	0D	2D	1
06	—	0	0E	11	1
07	—	0	0F	0D	1

(b) Page table: Only the first 16 PTEs are shown

Idx	Tag	Valid	Blk 0	Blk 1	Blk 2	Blk 3
0	19	1	99	11	23	11
1	15	0	—	—	—	—
2	1B	1	00	02	04	08
3	36	0	—	—	—	—
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	—	—	—	—
7	16	1	11	C2	DF	03
8	24	1	3A	00	51	89
9	2D	0	—	—	—	—
A	2D	1	93	15	DA	3B
B	0B	0	—	—	—	—
C	12	0	—	—	—	—
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	—	—	—	—

(c) Cache: Sixteen sets, 4-byte blocks, direct mapped

Select one:

1. PA = 0x0A28, PPN=0x20, PPO=0x20, CO=0, CI=0x08, CT=0x28, Cache Hit?= N, Byte = ...(Mem)

2. PA = 0x028, PPN=0xA20, PPO=0x20, CO=0, CI=0x08, CT=0x28, Cache Hit?= N, Byte = ...(Disk)

3. PA = 0x0A20, PPN=0x28, PPO=0x0, CO=0x20, CI=0x28, CT=0x08, Cache Hit?= N, Byte = ...(Mem)

4. PA = 0x0A20, PPN=0x28, PPO=0x20, CO=0, CI=0x08, CT=0x28, Cache Hit?= N, Byte = ...(Mem)

The correct answer is: PA = 0x0A20, PPN=0x28, PPO=0x20, CO=0, CI=0x08, CT=0x28, Cache Hit?= N, Byte = ...(Mem)

Question **75**

Not answered

Marked out of 5.00

A ………………………… models the execution of n concurrent threads as a trajectory through an n-dimensional Cartesian space.

Select one:

- 1. process graph
- 2. Bar graph
- 3. Pie graph
- 4. progress graph

The correct answer is: progress graph

Question **76**

Not answered

Marked out of 5.00

……………………… is a form of concurrent programming where applications explicitly schedule their own logical flows in the context of a single process.

Select one:

- 1. Process
- 2. I/O Multiplexing
- 3. Thread
- 4. Inter-process communication (IPC)

The correct answer is: I/O Multiplexing

Question **77**

Not answered

Marked out of 10.00

Consider the following program where main thread creates 5 threads.

Identify all the local static variables in the following program.

```
char **ptr1;
char **ptr2;
int  count;

int main()
{
   long i;
   pthread_t tid[5];

   char *msgs1[2] = {
      "Hello from thread0",
      "Hello from thread1"
   };
  char *msgs2[3] = {
      "Hello from thread3",
      "Hello from thread4",
       "Hello from thread5"
   };
   count=100;
   ptr1 = msgs1;
   ptr2 = msgs2;

   for (i = 0; i < 2; i++)
      pthread_create(&tid[i],  NULL, thread1, (void *)i);
  for (i = 2; i < 5; i++)
      pthread_create(&tid[i],  NULL, thread2, (void *)i);
   printf("THREAD MAIN EXITING\n");
   Pthread_exit(NULL);
}
```

```
void *thread1(void *vargp1)
{
long myid1 = (long)vargp1;
 static int cnt1 = 0;

count *= ++cnt1;
printf("[%ld]:  %s (cnt=%d) – count=%d \n",
        myid1, ptr1[myid1], cnt1, count);
 printf("THREAD %ld RETURNING \n", myid1);
 return NULL;
}

void *thread2(void *vargp2)
{
long myid2 = (long)vargp2;
static int cnt2 = 3;

count *= cnt2--;
printf("[%ld]:  %s (cnt=%d) – count=%d \n", myid2,
ptr2[myid2-2], cnt2, count);
printf("THREAD %ld RETURNING \n", myid2);
return NULL;
}
```

Select one:

- 1. thread1:cnt1, myid1, thread2:cnt2, myid1
- 2. main:ptr1, ptr2, count, thread1:cnt1,count, thread2:cnt2, count
- 3. main:ptr1, ptr2, count, thread1:cnt1, thread2:cnt2
- 4. thread1:cnt1, thread2:cnt2

The correct answer is: thread1:cnt1, thread2:cnt2

Question 78
Not answered
Marked out of 5.00

An example of Synchronous exception fault is ....

Select one:

1. Parity error
2. Illegal Instruction
3. Page fault
4. System Call

The correct answer is: Page fault

Question 79
Not answered
Marked out of 5.00

If any parent process terminates without reaping a child, then the orphaned child will be reaped by ....

Select one:

1. concurrent process
2. process scheduler
3. swapper process
4. init process

The correct answer is: init process

Question 80
Not answered
Marked out of 5.00

One similarity between threads and processes ....

Select one:

1. Both thread & process do not share code and data
2. Both thread & process are less expensive.
3. Both thread & process are not context switched
4. Both threads and processes can run concurrently with others (possibly on different cores)

The correct answer is: Both threads and processes can run concurrently with others (possibly on different cores)

Question 81
Not answered
Marked out of 10.00

Consider a virtual memory system having a 48-bit virtual address space partitioned into 4KB pages.

Determine the number of bits in VPN and VPO.

Select one:

1. VPN=40 bits VPO=8 bits
2. VPN=24 bits VPO=24 bits
3. VPN=36 bits VPO=12 bits
4. VPN=12 bits VPO=36 bits

The correct answer is: VPN=36 bits VPO=12 bits

Question 82
Not answered
Marked out of 10.00

Unix provides a system call called .... that allows application programs to do their own memory mapping.

Answer:

The correct answer is: mmap

Question 83
Not answered
Marked out of 6.00

Virtual address is divided into two parts :............., ...

Select one:
1. PPN, PPO
2. VPN, VPO
3. VPA, PPA
4. LPA, RPA

The correct answer is: VPN, VPO

Question 84
Not answered
Marked out of 5.00

In UNIX/Linux, all input and output operations are performed by reading and writing the appropriate ………..

Answer:

The correct answer is: files

Question 85
Not answered
Marked out of 5.00

The POSIX library function to terminate a thread is ……………………….

Select one:
1. pthread_exit
2. pthread_detach
3. pthread_cancel
4. pthread_join

The correct answer is: pthread_exit

Question 86
Not answered
Marked out of 10.00

setjmp function initially returns a value …………………. and after that it returns the value provided by the function ………………………………………..

Select one:

1. 0, longjmp
2. 1, longjmp
3. 1, fork
4. 0, execve

The correct answer is: 0, longjmp

Question 87
Not answered
Marked out of 5.00

The default action taken by the kernel after a process receives the signal SIGSTOP is……………..

Select one:

1. Terminate
2. Terminate with Dump
3. Suspend
4. Ignore

The correct answer is: Suspend

Question 88
Not answered
Marked out of 10.00

The ………………. operations in P occur indivisibly, in the sense that once the semaphore s becomes nonzero, the decrement of s occurs without interruption.

Answer: ✗

The correct answer is: test and decrement

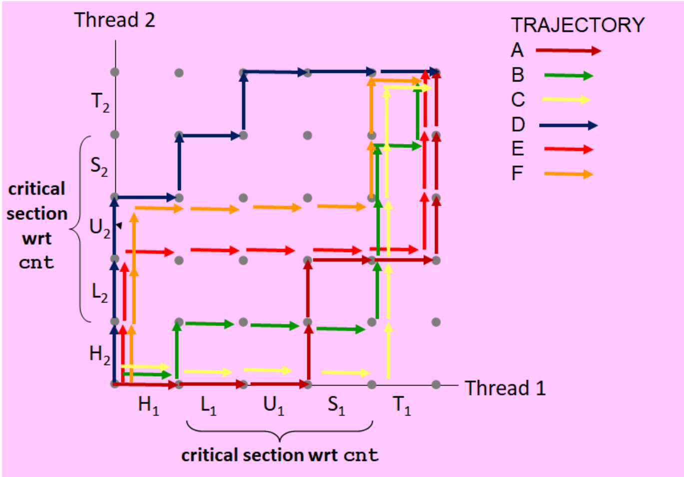
Question 89

Not answered

Marked out of 30.00

For the progress graph given in Figure , classify the following trajectories

A, B, C, D, E, F as unsafe trajectories.



Select one:

1. A, B, C
2. A, E, F
3. D, E, F
4. B, C, D

The correct answer is: A, E, F

Question 90

Not answered

Marked out of 10.00

.... Returns the exit status of a normally terminated child.

Select one:

1. WIFEXITED(status)
2. WIFSIGNALED(status)
3. WEXITSTATUS(status)
4. WTERMSIG(status)

The correct answer is: WEXITSTATUS(status)

Question 91

Not answered

Marked out of 5.00

If any parent terminates without reaping a child, then the orphaned child will be reaped by .... process.

Answer:

The correct answer is: init

Question 92
Not answered
Marked out of 30.00

Consider the following program where main thread creates 5 threads.
Identify all the local variables in the following program.

```
char **ptr1;
char **ptr2;
int  count;

int main()
{
   long i;
   pthread_t tid[5];

   char *msgs1[2] = {
      "Hello from thread0",
      "Hello from thread1"
   };
  char *msgs2[3] = {
      "Hello from thread3",
      "Hello from thread4",
       "Hello from thread5"
   };
   count=100;
   ptr1 = msgs1;
   ptr2 = msgs2;

   for (i = 0; i < 2; i++)
      pthread_create(&tid[i],  NULL, thread1, (void *)i);
  for (i = 2; i < 5; i++)
      pthread_create(&tid[i],  NULL, thread2, (void *)i);
   printf("THREAD MAIN EXITING\n");
   Pthread_exit(NULL);
}
```

```
void *thread1(void *vargp1)
{
long myid1 = (long)vargp1;
 static int cnt1 = 0;

count *= ++cnt1;
printf("[%ld]:  %s (cnt=%d) – count=%d \n",
        myid1, ptr1[myid1], cnt1, count);
 printf("THREAD %ld RETURNING \n", myid1);
 return NULL;
}

void *thread2(void *vargp2)
{
long myid2 = (long)vargp2;
static int cnt2 = 3;

count *= cnt2--;
printf("[%ld]:  %s (cnt=%d) – count=%d \n", myid2,
ptr2[myid2-2], cnt2, count);
printf("THREAD %ld RETURNING \n", myid2);
return NULL;
}
```

Select one:

1. main: i, tid, msg1, msg2, thread1: cnt1, thread2: cnt2
2. main: i, tid, msg1, msg2, thread1: myid1, thread2: myid2
3. main: i, tid, msg1, msg2,ptr1, ptr2, thread1: myid1, count, thread2: myid2, count
4. main: i, tid, msg1, msg2 thread1: cnt1, count, thread2: cnt2, count

The correct answer is: main: i, tid, msg1, msg2, thread1: myid1, thread2: myid2

Question 93
Not answered
Marked out of 5.00

The ID number of x86-64 System call read is ·········

Select one:

1. 3
2. 1
3. 0
4. 2

The correct answer is: 0

Question 94
Not answered
Marked out of 5.00

One difference between thread and process ：................

Select one:
- 1. Both threads and processes can run concurrently with others (possibly on different cores)
- 2. Both thread & process are context switched
- 3. Both thread & process have their own logical control flows
- 4. Threads share all code and data (except local stacks) & Processes (typically) do not.

The correct answer is: Threads share all code and data (except local stacks) & Processes (typically) do not.

Question 95
Not answered
Marked out of 5.00

The default action taken by the kernel after a process receives the signal SIGKILL is……………….,.

Select one:
- 1. Terminate with Dump
- 2. Ignore
- 3. Terminate
- 4. Suspend

The correct answer is: Terminate

Question 96
Not answered
Marked out of 5.00

A variable x is ……………if and only if multiple threads reference some instance of x.

Select one:
- 1. local
- 2. dynamic
- 3. static
- 4. shared

The correct answer is: shared

Question 97
Not answered
Marked out of 5.00

A .... is a logical flow that runs in the context of a process.

Answer:

The correct answer is: thread

Question 98
Not answered
Marked out of 10.00

Identify the correct P function ....

Select one:

1. void P(sem_t *s)

   {
     sem_post(&s);
   }
2. void P(sem_t *s)
   {
     sem_decr(&s);
   }
3. void P(sem_t *s)
   {
     sem_test(&s);
   }
4. void P(sem_t *s)
   {
     sem_wait(&s);
   }

The correct answer is: void P(sem_t *s)
{
  sem_wait(&s);
}

Question 99
Not answered
Marked out of 5.00

If a child process terminates after the parent process dies then the child process is termed as ·······················..

Select one:

1. Dead process
2. Bad process
3. Orphan process
4. Zombie process

The correct answer is: Orphan process

Question 100
Not answered
Marked out of 15.00

Short counts can occur in these situations:

Select one or more:

1. Writing to disk files
2. Reading and writing network sockets
3. Reading text lines from a terminal
4. Reading from disk files (except for EOF)
5. Encountering (end-of-file) EOF on reads

The correct answers are: Encountering (end-of-file) EOF on reads, Reading text lines from a terminal, Reading and writing network sockets

Question 101
Not answered
Marked out of 10.00

Consider a virtual memory system having a 64-bit virtual address space partitioned into 4KB pages.

Determine the number of bits in VPN and VPO

Select one:
1. VPN=52 bits   VPO=12 bits
2. VPN=32 bits   VPO=32 bits
3. VPN=12 bits   VPO=52 bits
4. VPN=56 bits   VPO=8 bits

The correct answer is: VPN=52 bits   VPO=12 bits

Question 102
Not answered
Marked out of 10.00

A process is .... iff it is waiting for a condition that will never be true.

Answer:

The correct answer is: deadlocked

Question 103
Not answered
Marked out of 5.00

If a child process terminates before the parent process dies then the child process is termed as ……………………..

Select one:
1. Bad process
2. Zombie process
3. Dead process
4. Orphan process

The correct answer is: Zombie process

Question 104
Not answered
Marked out of 10.00

Event-driven graphical user interface is an example of ....

Select one:
1. Travelling Salesman Problem
2. Producer-Consumer Problem
3. Dining Philosophers Problem
4. Readers-Writers Problem

The correct answer is: Producer-Consumer Problem

Question 105
Not answered
Marked out of 10.00

The overlapping forbidden regions induce a set of states called the .... region.

Answer: ✗

The correct answer is: deadlock

Question 106
Not answered
Marked out of 5.00

If an interrupt is generated from outside the CPU then it is termed as ....

Select one:
1. unacceptable
2. asynchronous
3. unhandleable
4. synchronous

The correct answers are: asynchronous , synchronous

Question 107
Not answered
Marked out of 5.00

The file descriptor associated with standard error (stderr) is ....

Select one:
1. 1
2. 3
3. 2
4. 0

The correct answer is: 2

Question 108
Not answered
Marked out of 5.00

If the required page is found in the main memory then it is termed as ....

Select one:
1. Page Fault
2. Cache Miss
3. Cache hit
4. Page Hit

The correct answer is: Page Hit

Question 109
Not answered
Marked out of 5.00

The read function returns ..... on encountering error.

Select one:
1. 0
2. -1
3. number of bytes read
4. 1

The correct answer is: -1

Question 110
Not answered
Marked out of 10.00

Consider a virtual memory system having a 48-bit virtual address space partitioned into 4MB pages.

Determine the number of bits in VPN and VPO

Select one:
1. VPN=26 bits   VPO=22 bits
2. VPN=24 bits   VPO=24 bits
3. VPN=40 bits   VPO=8 bits
4. VPN=22 bits   VPO=26 bits

The correct answer is: VPN=26 bits   VPO=22 bits

Question 111
Not answered
Marked out of 5.00

The .... command is used to get back to the beginning of the file.

Select one:
1. lseek(fd, 100L, 0);
2. lseek(fd, 0L, 1);
3. lseek(fd, 0L, 2);
4. lseek(fd, 0L, 0);

The correct answer is: lseek(fd, 0L, 0);

Question 112
Not answered
Marked out of 5.00

An example of Synchronous exception Abort is ....

Select one:
1. Floating point exception
2. System Call
3. parity error
4. Page fault

The correct answer is: parity error

Question 113
Not answered
Marked out of 10.00

Given a Virtual Memory System having 48-bit virtual address, 64KB Page size and using a single Page table,

Determine the number of entries in the page table(i.e., number of PTEs).

Select one:
1. PTEs = 4Giga entries
2. PTEs = 4Mega entries
3. PTEs = 2Giga entries
4. PTEs = 2Mega entries

The correct answer is: PTEs = 4Giga entries

Question 114
Not answered
Marked out of 5.00

The threads are scheduled automatically by the...... .

Answer:

The correct answer is: kernel

Question 115
Not answered
Marked out of 5.00

The ID number of x86-64 System call kill is ·········

Select one:
1. 59
2. 57
3. 60
4. 62

The correct answer is: 62

Question 116
Not answered
Marked out of 10.00

Multimedia processing is an example of ....

Select one:
1. Producer-Consumer Problem
2. Travelling Salesman Problem
3. Dining Philosophers Problem
4. Readers-Writers Problem

The correct answer is: Producer-Consumer Problem

Question 117
Not answered
Marked out of 10.00

Each process created by a Unix shell begins life with the following open files....................

Select one:

1. standard input (stdin)
2. standard output (stdout)
3. standard error (stderr)
4. All of the above

The correct answer is: All of the above

Question 118
Not answered
Marked out of 10.00

If the waitpid function was interrupted by a signal, then returns ..…… and set errno to …………

Select one:

1. 0, EINTR
2. -1, ECHILD
3. -1, EINTR
4. 0, ECHILD

The correct answer is: -1, EINTR

Question 119
Not answered
Marked out of 5.00

The ID number of x86-64 System call close is ………

Select one:

1. 3
2. 2
3. 0
4. 1

The correct answer is: 3

Question 120

Not answered

Marked out of 25.00

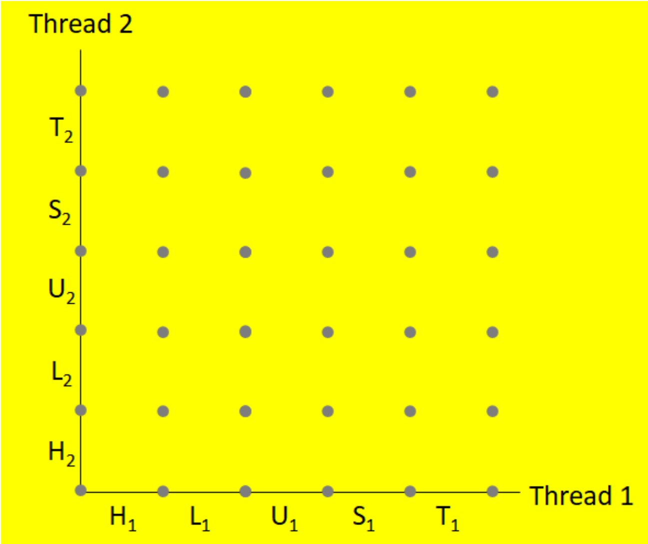
Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Draw the Progress Graph and show how the concurrent execution of the assembly code corresponding to the loop executed concurrently by the threads thread1 and thread2

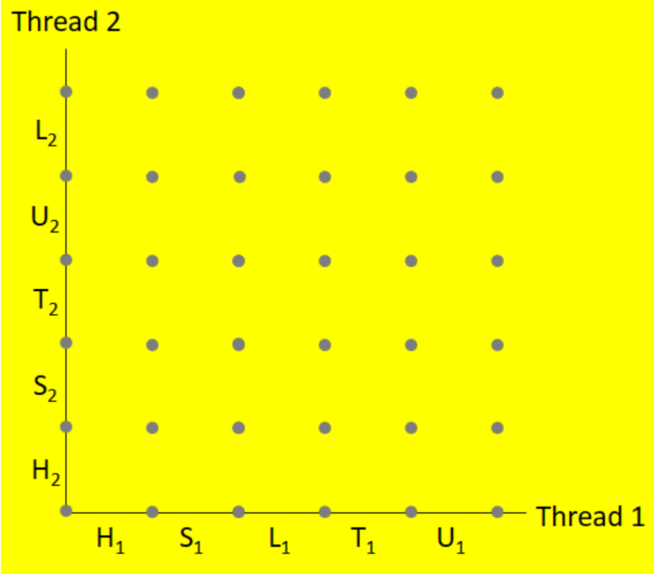
```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

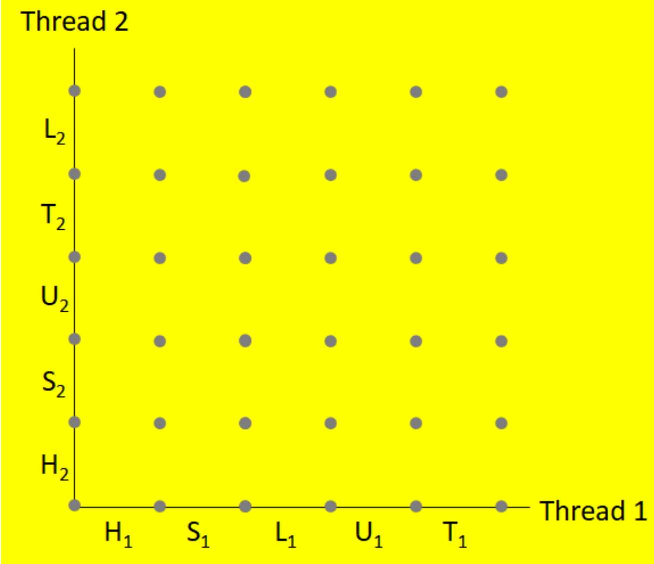
1.



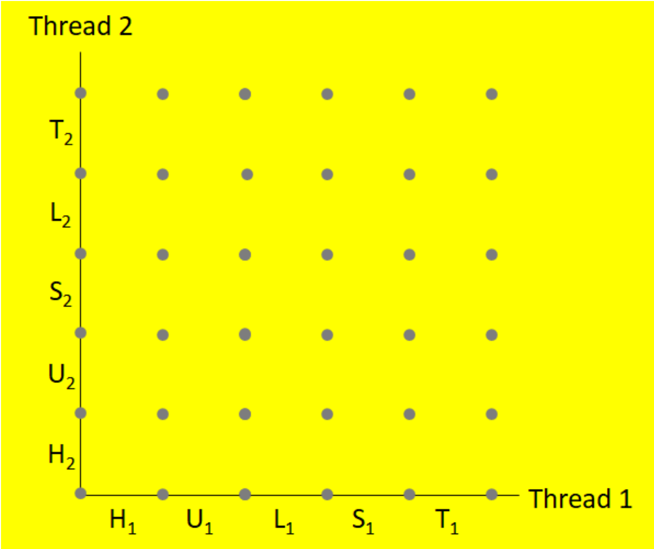
2.



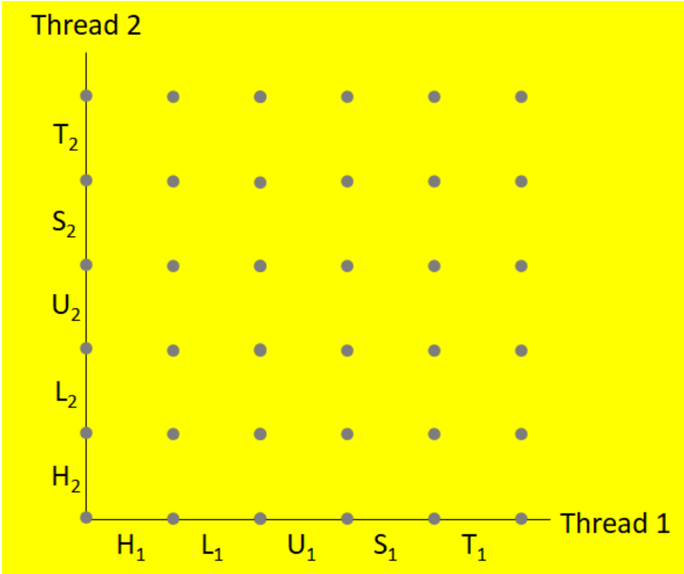
3.



4.



The correct answer is:



Question 121

Not answered

Marked out of 5.00

End of line (EOL) indicator used in Linux regular file is ……………………..

Select one:

1. Line Feed(LF)
2. Carriage Return followed by Line Feed
3. Carriage Return(CR)
4. Line Feed followed by Carriage Return

The correct answer is: Line Feed(LF)

Question 122

Not answered

Marked out of 30.00

Given the following program, Draw the process graph and Identify atleast one feasible output

```
int main()
{
        pid_t pid;
         printf("COVID 19 CORONA VIRUS ");
         if (fork() == 0 )
          {   printf("BE SAFE & HEALTHY");
                   if (Fork() != 0)
                         printf("PANDEMIC DISEASE");
          }
         if (Fork() !=0)
               printf("MAINTAIN DISTANCING");
         printf("STAY HOME \n");
}
```

Select one:

1. COVID 19 CORONA VIRUS
PANDEMIC DISEASE
MAINTAN DISTANCING
BE SAFE & HEALTHY
STAY HOME
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME
MAINTAIN DISTANCING
STAY HOME

2. COVID 19 CORONA VIRUS
MAINTAN DISTANCING
BE SAFE & HEALTHY
STAY HOME
STAY HOME
PANDEMIC DISEASE
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME
STAY HOME
MAINTAIN DISTANCING

3. COVID 19 CORONA VIRUS
PANDEMIC DISEASE
MAINTAN DISTANCING
BE SAFE & HEALTHY
STAY HOME
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME
STAY HOME
MAINTAIN DISTANCING

4. COVID 19 CORONA VIRUS
BE SAFE & HEALTHY
MAINTAN DISTANCING
STAY HOME
PANDEMIC DISEASE
STAY HOME
MAINTAIN DISTANCING
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME

The correct answer is: COVID 19 CORONA VIRUS
BE SAFE & HEALTHY
MAINTAN DISTANCING
STAY HOME
PANDEMIC DISEASE
STAY HOME
MAINTAIN DISTANCING
STAY HOME
MAINTAIN DISTANCING
STAY HOME
STAY HOME
STAY HOME

Question 123
Not answered
Marked out of 5.00

An example of Synchronous exception fault is ....

Select one:
- 1. Parity error
- 2. Illegal Instruction
- 3. System Call
- 4. Floating point exception

The correct answer is: Floating point exception

Question 124
Not answered
Marked out of 5.00

One difference between thread and process ....

Select one:
- 1. Both thread & process have their own logical control flows
- 2. Both threads and processes can run concurrently with others (possibly on different cores)
- 3. Threads are somewhat less expensive than processes
- 4. Both thread & process are context switched

The correct answer is: Threads are somewhat less expensive than processes

Question 125
Not answered
Marked out of 20.00

For the following program given: Draw the process graph and Write the outputs of both parent and child processes.

```
int main()
{
    pid_t pid;
    int x = 1;
    pid = Fork();
    if (pid == 0) {
        x=x+100;
        printf("child : x=%d\n", --x);
        exit(0);
    }
    x=x-100;
    printf("parent: x=%d\n", ++x);
    exit(0);
}
```

Select one:

1. parent: x=-99
   child : x=101
2. parent: x=-98
   child : x=100
3. parent: x=-100
   child : x=100
4. parent: x=2
   child : x=100

The correct answer is: parent: x=-98
child : x=100

Question 126
Not answered
Marked out of 5.00

………… is a Standard interface for ~60 functions that manipulate threads from C programs.

Answer:

The correct answer is: Pthreads

Question 127
Not answered
Marked out of 5.00

The ID number of x86-64 System call open is ………

Select one:

1. 0
2. 3
3. 2
4. 1

The correct answer is: 2

Question 128

Not answered

Marked out of 5.00

The POSIX library function to determine its own thread ID by a calling thread is ………………………

Select one:

1. pthread_detach
2. pthread_self
3. pthread_join
4. pthread_create

The correct answer is: pthread_self

Question 129

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one infeasible output

```
Int main()
{
 printf("L0\n");
   if (fork() != 0) {
       printf("L1\n");
       if (fork() != 0) {
           printf("L2\n");
             }
   }
   printf("Bye\n");
}
```

Select one:

1. L0
L1
Bye
Bye
L2
Bye

2. L0
Bye
L1
Bye
Bye
L2

3. L0
Bye
L1
Bye
L2
Bye

4. L0
L1
Bye
L2
Bye
Bye

The correct answer is: L0
Bye
L1
Bye
Bye
L2

Question 130
Not answered
Marked out of 5.00

……………………… are logical flows that run in the context of a single process and are scheduled by the kernel.

Select one:
1. I/O Multiplexing
2. Threads
3. Processes
4. Inter-process communication (IPC)

The correct answer is: Threads

Question 131
Not answered
Marked out of 10.00

Function longjmp is …………………………………….

Select one:
1. called once, returns twice
2. called once, returns only once
3. called once, returns one or more times
4. called once, never returns

The correct answer is: called once, never returns

Question 132
Not answered
Marked out of 5.00

Process = ... + code, data, and stack

Select one:
1. process context
2. Kernel context
3. Program context
4. Thread Context

The correct answer is: process context

Question 133
Not answered
Marked out of 5.00

The default action taken by the kernel after a process receives the signal SIGINT is………………..

Select one:
1. Suspend
2. Terminate with Dump
3. Ignore
4. Terminate

The correct answer is: Terminate

Question 134
Not answered
Marked out of 5.00

………………………… is a condition where a set of processes are waiting because each process is holding on to a resource and waiting for some other resource that is being held by another process which is also waiting for the resource held by another resource in a chain. Hence, no process can go forward as no process is going to release the resource held by it.

Select one:
- 1. Race condition
- 2. Live lock
- 3. Deadlock
- 4. Starvation

The correct answer is: Deadlock

Question 135
Not answered
Marked out of 10.00

The main impact of this notion of a .... is that a thread can kill any of its peers or wait for any of its peers to terminate. Further, each peer can read and write the same shared data.

Answer:

The correct answer is: pool of peers

Question 136
Not answered
Marked out of 10.00

Since processes have separate virtual address spaces, flows that want to communicate with each other must use some kind of explicit ………………………………mechanism.

Select one:
- 1. Parameter Passing
- 2. Function call & return
- 3. Inter-process communication (IPC)
- 4. Modular Programming

The correct answer is: Inter-process communication (IPC)

Question 137

Not answered

Marked out of 25.00

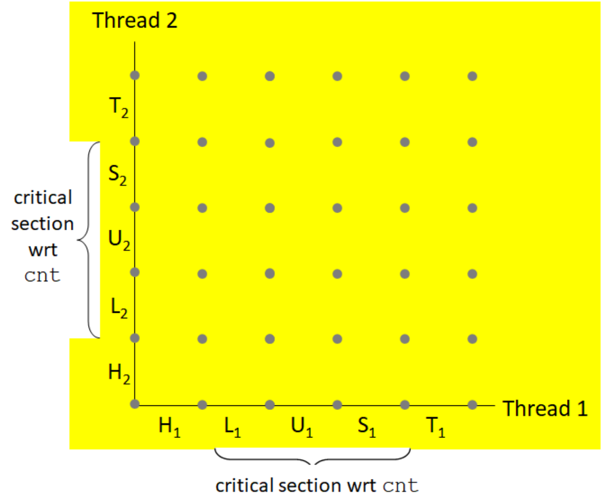
Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Draw the Progress Graph and Identify the critical region.

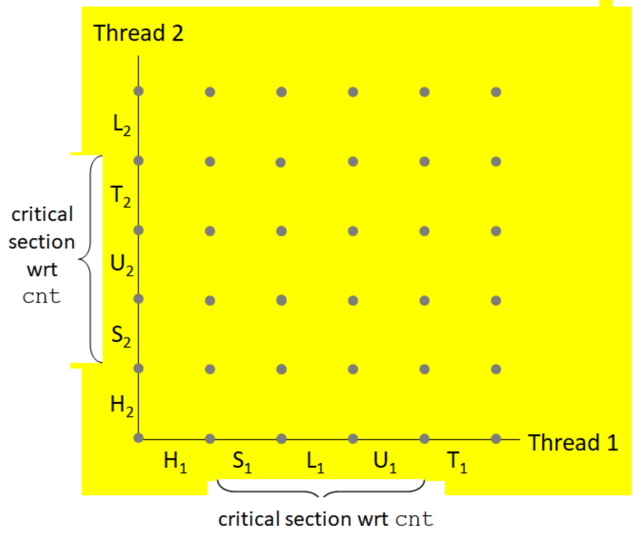
```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

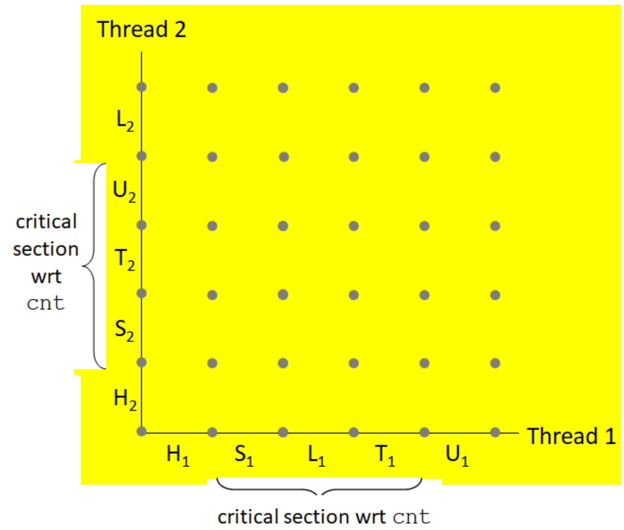
1.



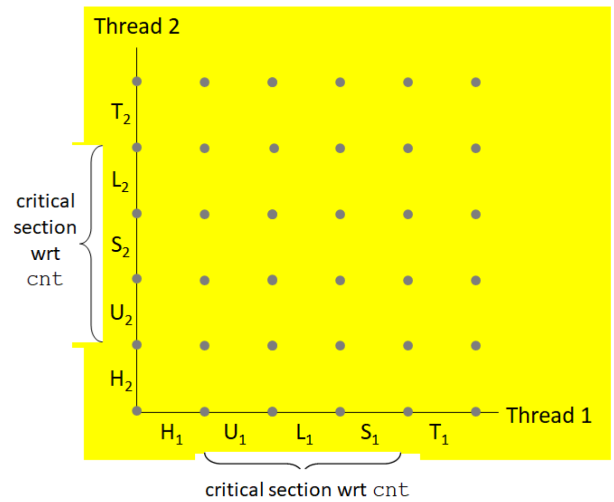
2.



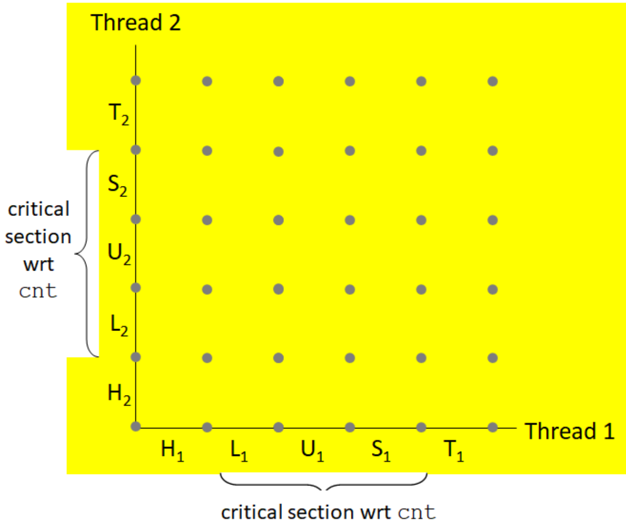
3.



4.



The correct answer is:



Question 138
Not answered
Marked out of 5.00

TLB stands for ···.

Select one:
- 1. Transformation Look-alike Buffer
- 2. Translation Look-alike Buffer
- 3. Translation Look-aside Buffer
- 4. Transformation Look-aside Buffer

The correct answer is: Translation Look-aside Buffer

Question 139
Not answered
Marked out of 10.00

Another important use of ...., besides providing mutual exclusion, is to schedule accesses to shared resources.

Answer: ✗

The correct answer is: semaphores

Question 140
Not answered
Marked out of 5.00

In UNIX/LINUX, All I/O devices, such as networks, disks, and terminals, are modeled as ·········

Answer: ✗

The correct answer is: files

Question 141
Not answered
Marked out of 5.00

In some situations, read and write transfer fewer bytes than the application requests. This problem is known as ....

Select one:
- 1. Short Counts
- 2. Buffer Overflow
- 3. Buffer Underflow
- 4. File Flush

The correct answer is: Short Counts

Question 142
Not answered
Marked out of 5.00

One similarity between thread and process : ....

Select one:
- 1. Both thread & process do not share code and data
- 2. Both thread & process are less expensive.
- 3. Both thread & process are not context switched
- 4. Both thread & process have their own logical control flows

The correct answer is: Both thread & process have their own logical control flows

Question 143
Not answered
Marked out of 5.00

The POSIX library function to make a peer thread non-joinable is ....

Select one:
- 1. pthread_exit
- 2. pthread_join
- 3. pthread_detach
- 4. pthread_cancel

The correct answer is: pthread_detach

Question 144
Not answered
Marked out of 10.00

The .... function loads and runs a new program in the context of the current process.

Answer: ✗

The correct answer is: execve

Question 145
Not answered
Marked out of 10.00

Identify the basic approaches for building concurrent programs provided by modern operating systems :

Select one:
- 1. Processes
- 2. I/O Multiplexing
- 3. Threads
- 4. All of the above

The correct answer is: All of the above

Question 146
Not answered
Marked out of 5.00

Internally generated interrupts that occur as a result of executing instructions are termed as ....

Answer:

The correct answer is: exceptions

Question 147
Not answered
Marked out of 5.00

Identify Class of thread-unsafe functions: ....

Select one:
- 1. Functions that protect shared variables
- 2. Functions that do not call thread-unsafe functions
- 3. Functions that do not return a pointer to a static variable
- 4. Functions that keep state across multiple invocations

The correct answer is: Functions that keep state across multiple invocations

Question 148
Not answered
Marked out of 10.00

FULL FORM OF PCI ....

Answer:

The correct answer is: Peripheral Component Interconnect

Question 149
Not answered
Marked out of 5.00

The .... command is used to seek to the end of the file.

Select one:
- 1. lseek(fd, 0L, 2);
- 2. lseek(fd, 0L, 1);
- 3. lseek(fd, 100L, 0);
- 4. lseek(fd, 0L, 0);

The correct answer is: lseek(fd, 0L, 2);

Question 150
Not answered
Marked out of 5.00

A ················is a small message that notifies a process that an event of some type has occurred in the system.

Select one:
- 1. Signal
- 2. Exception
- 3. System Call
- 4. Interrupt

The correct answer is: Signal

Question 151
Not answered
Marked out of 10.00

In order to avoid ...., each joinable thread should either be explicitly reaped by another thread, or detached by a call to the pthread_detach function.

Answer:

The correct answer is: memory leaks

Question 152
Not answered
Marked out of 5.00

························ occurs when all the processes are allowed to access the shared resource (critical section) at the same time thereby producing an incorrect result

Select one:
- 1. Race condition
- 2. Deadlock
- 3. Live lock
- 4. Starvation

The correct answer is: Race condition

Question 153
Not answered
Marked out of 5.00

Synchronous exception Faults are ……………………………………………………………………

Select one:
- 1. Unintentional & Unrecoverable
- 2. Intentional & Unrecoverable
- 3. Intentional
- 4. Unintentional but possibly recoverable

The correct answer is: Unintentional but possibly recoverable

Question 154
Not answered
Marked out of 5.00

The default action taken by the kernel after a process receives the signal SIGQUIT is………………..

Select one:
- 1. Ignore
- 2. Suspend
- 3. Terminate
- 4. Terminate with Dump

The correct answer is: Terminate with Dump

Question 155
Not answered
Marked out of 5.00

………………… function is used by Linux process to delete regions of virtual memory.

Select one:
- 1. pmap
- 2. munmap
- 3. mmap
- 4. vmap

The correct answer is: munmap

Question 156
Not answered
Marked out of 5.00

Applications can explicitly block and unblock selected signals using the …………………………function:

Select one:
- 1. sigfillset
- 2. sigaddset
- 3. sigdelset
- 4. sigprocmask

The correct answer is: sigprocmask

Question 157

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and Write the outputs of both parent and child processes.

```
int main()
  {
     fork();
     fork();
     printf("hello\n");
     exit(0);
  }
```

Select one:

1. hello
hello

2. hello
hello
hello

3. hello

4. hello
hello
hello
hello

The correct answer is: hello
hello
hello
hello

Question 158

Not answered

Marked out of 20.00

For the following program given: Draw the process graph and Write the statements executed by only the child process.

```
int main()
{
    pid_t pid;
    int x = 1;
    pid = Fork();
    if (pid == 0) {
        x=x+100;
        printf("child : x=%d\n", --x);
        exit(0);
    }
    x=x-100;
    printf("parent: x=%d\n", ++x);
    exit(0);
}
```

Select one:

1.

```
pid = Fork();
x=x+100;
printf("child : x=%d\n", --x);
exit(0);
```

2.

```
pid = Fork();
x=x+100;
printf("child : x=%d\n", --x);
x=x-100 ;
printf("parent: x=%d\n", ++x);
exit(0);
```

3.

```
x=x+100;
printf("child : x=%d\n", --x);
x=x-100 ;
printf("parent: x=%d\n", ++x);
exit(0);
```

4.

```
x=x+100;
printf("child : x=%d\n", --x);
exit(0);
```

The correct answer is:

```
x=x+100;
printf("child : x=%d\n", --x);
exit(0);
```

Question 159

Not answered

Marked out of 20.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 24 bits wide.

Physical addresses(PA) are 18 bits wide.

The page size is 1024 bytes.

The TLB is four-way set associative with 32 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 8-byte line(block)size and 64 total blocks.

Cache block is byte addressable.

Determine the number of bits in the following

VPN = ……..bits, VPO = ……bits, TLBI = ……..bits, TLBT = …… bits

Total Number of entries in Page Table, PTEs = …… entries

Select one:

1. VPN=14 bits, VPO=10 bits, TLBI=3 bits, TLBT=11 bits, PTEs=32K entries

2. VPN=10 bits, VPO=14 bits, TLBI=3 bits, TLBT=11 bits, PTEs=16K entries

3. VPN=14 bits, VPO=10 bits, TLBI=11 bits, TLBT=3 bits, PTEs=16K entries

4. VPN=14 bits, VPO=10 bits, TLBI=3 bits, TLBT=11 bits, PTEs=16K entries

The correct answer is: VPN=14 bits, VPO=10 bits, TLBI=3 bits, TLBT=11 bits, PTEs=16K entries

Question 160

Not answered

Marked out of 15.00

Reorder the following steps in the correct order to indicate address translation performed by the on-chip MMU when there is a TLB hit.

a) The MMU fetches the appropriate PTE from the TLB.

b) The MMU translates the virtual address to a physical address and sends it to the cache/main memory.

c) The CPU generates a virtual address.

d) The cache/main memory returns the requested data word to the CPU.

Select one:

1. c),b),a),d)

2. c),b),d),a)

3. a),b),c),d)

4. c),a),b),d)

The correct answer is: c),a),b),d)

Question 161

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one feasible output

```
int main()
{
   printf("L0\n");
   if (fork() == 0) {
      printf("L1\n");
      if (fork() == 0) {
         printf("L2\n");
      }
   }
   printf("Bye\n");
}
```

Select one:

1. L0
Bye
L1
L2
Bye
Bye

2. L0
Bye
L1
Bye
Bye
L2

3. L0
L1
Bye
Bye
Bye
L2

4. L0
Bye
Bye
L1
Bye
L2

The correct answer is: L0
Bye
L1
L2
Bye
Bye

Question 162

Not answered

Marked out of 5.00

Threads associated with process form a pool of ....

Answer:

The correct answer is: peers

Question 163
Not answered
Marked out of 5.00

Example of an asynchronous interrupt is ....

Select one:
- 1. timer interrupt
- 2. overflow interrupt
- 3. breakpoint interrupt
- 4. divide by zero interrupt

The correct answer is: timer interrupt

Question 164
Not answered
Marked out of 20.00

Given the following program, Draw the process graph and Determine the total number of child processes created when this program is executed

```
int main()
   {
       fork();
       fork();
       printf(“hello\n”);
       exit(0);
   }
```

Select one:
- 1. 3
- 2. 2
- 3. 8
- 4. 4

The correct answer is: 3

Question 165
Not answered
Marked out of 5.00

If the required page is not found in the main memory then it is termed as …………..

Select one:
- 1. Page Hit
- 2. Cache hit
- 3. Page Fault
- 4. Cache Miss

The correct answer is: Page Fault

Question 166
Not answered
Marked out of 5.00

Identify Class of thread-safe functions: ....

Select one:

1. Functions that return a pointer to a static variable
2. Functions that protect shared variables
3. Functions that call thread-unsafe functions
4. Functions that keep state across multiple invocations

The correct answer is: Functions that protect shared variables

Question 167
Not answered
Marked out of 20.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

Cache block is byte addressable.

Determine the number of bits in the following

VPN = ... bits, VPO = ... bits, TLBI = ... bits, TLBT = ... bits

Total Number of entries in Page Table, PTEs = ... entries

Select one:

1. VPN=8 bits, VPO=6 bits, TLBI=6 bits, TLBT=2 bits, PTEs=256 entries
2. VPN=6 bits, VPO=8 bits, TLBI=2 bits, TLBT=6 bits, PTEs=256 entries
3. VPN=8 bits, VPO=6 bits, TLBI=2 bits, TLBT=6 bits, PTEs=256 entries
4. VPN=8 bits, VPO=6 bits, TLBI=2 bits, TLBT=6 bits, PTEs=512 entries

The correct answer is: VPN=8 bits, VPO=6 bits, TLBI=2 bits, TLBT=6 bits, PTEs=256 entries

Question 168
Not answered
Marked out of 5.00

Processes form a .... hierarchy.

Answer:

The correct answer is: tree

Question 169
Not answered
Marked out of 5.00

Input/output (I/O) devices such as graphics cards, monitors, mice, keyboards, and disks are connected to the CPU and main memory using an I/O bus such as Intel's ....

Select one:
- 1. ISA BUS
- 2. THUNDERBOLT
- 3. MCA BUS
- 4. PCI BUS

The correct answer is: PCI BUS

Question 170
Not answered
Marked out of 10.00

Given a Virtual Memory System having 48-bit virtual address, 4KB Page size and using a single Page table, determine the number of bits in VPN and VPO fields of the virtual address.

Select one:
- 1. VPN=24 bits VPO= 24 bits
- 2. VPN=36 bits VPO= 12 bits
- 3. VPN=40 bits VPO= 8 bits
- 4. VPN=12 bits VPO= 36 bits

The correct answer is: VPN=36 bits VPO= 12 bits

Question 171
Not answered
Marked out of 20.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 24 bits wide.

Physical addresses(PA) are 18 bits wide.

The page size is 1024 bytes.

The TLB is four-way set associative with 32 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 8-byte line(block)size and 64 total blocks.

Cache block is byte addressable.

Determine the number of bits in the following

PPN = ... bits, PPO = ... bits, CO = ... bits, CI = ... bits, CT = ... bits

Select one:
- 1. PPN=10 bits, PPO=8 bits, CO=3 bits, CI=4 bits, CT=11 bits
- 2. PPN=8 bits, PPO=10 bits, CO=4 bits, CI=3 bits, CT=11 bits
- 3. PPN=8 bits, PPO=10 bits, CO=2 bits, CI=3 bits, CT=13 bits
- 4. PPN=8 bits, PPO=10 bits, CO=3 bits, CI=4 bits, CT=11 bits

The correct answer is: PPN=8 bits, PPO=10 bits, CO=3 bits, CI=4 bits, CT=11 bits

Question 172
Not answered
Marked out of 5.00

………………………… occurs when two or more processes continually repeat the same interaction in response to changes in the other processes without doing any useful work. These processes are not in the waiting state, and they are running concurrently without progressing forward.

Select one:

1. Live lock
2. Race condition
3. Deadlock
4. Starvation

The correct answer is: Live lock

Question 173
Not answered
Marked out of 5.00

……………… simplifies memory management by providing each process with an uniform address space.

Select one:

1. Fully Associative Cache
2. Virtual Memory
3. Set Associative Cache
4. Direct Mapped Cache

The correct answer is: Virtual Memory

Question 174

Not answered

Marked out of 30.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

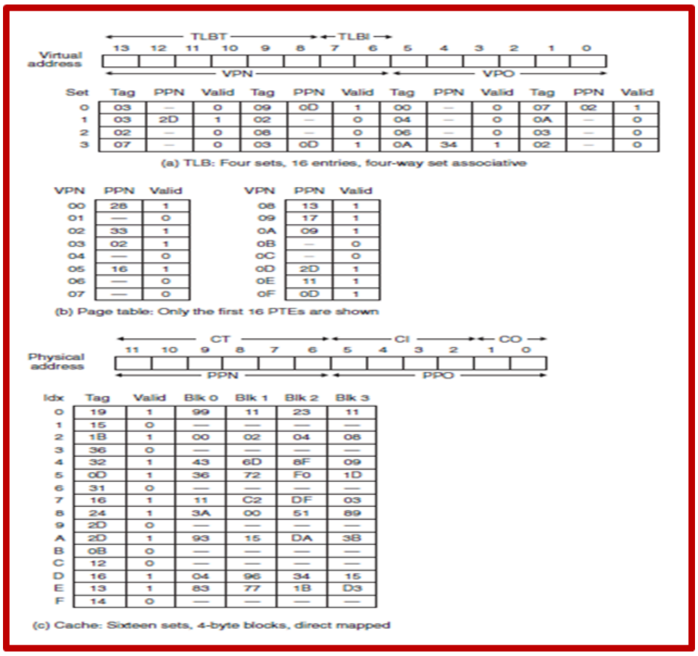
Cache block is byte addressable.

Processor generates a virtual address VA = 0x03D4, Translate this to physical address PA using TLB or Page Table

Determine the values (in hex) in the following fields:

Physical Address PA = ..., PPN = ..., PPO= .....

CO=....., CI=......, CT=......, Cache Hit?.... (Y/N), Byte (read from cache)=......



Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	—	0	09	0D	1	00	—	0	07	02	1
1	03	2D	1	02	—	0	04	—	0	0A	—	0
2	02	—	0	08	—	0	06	—	0	03	—	0
3	07	—	0	03	0D	1	0A	34	1	02	—	0

(a) TLB: Four sets, 16 entries, four-way set associative

VPN	PPN	Valid	VPN	PPN	Valid
00	28	1	08	13	1
01	—	0	09	17	1
02	33	1	0A	09	1
03	02	1	0B	—	0
04	—	0	0C	—	0
05	16	1	0D	2D	1
06	—	0	0E	11	1
07	—	0	0F	0D	1

(b) Page table: Only the first 16 PTEs are shown

Idx	Tag	Valid	Blk 0	Blk 1	Blk 2	Blk 3
0	19	1	99	11	23	11
1	15	0	—	—	—	—
2	1B	1	00	02	04	08
3	36	0	—	—	—	—
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	—	—	—	—
7	16	1	11	C2	DF	03
8	24	1	3A	00	51	89
9	2D	0	—	—	—	—
A	2D	1	93	15	DA	3B
B	0B	0	—	—	—	—
C	12	0	—	—	—	—
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	—	—	—	—

(c) Cache: Sixteen sets, 4-byte blocks, direct mapped

Select one:

- 1. PA = 0x0254, PPN=0x0D, PPO=0x14, CO=0, CI=0x05, CT=0x0D, Cache Hit?= Y, Byte =0x26
- 2. PA = 0x0354, PPN=0x0D, PPO=0x14, CO=0, CI=0x05, CT=0x0D, Cache Hit?= Y, Byte =0x36
- 3. PA = 0x0354, PPN=0x14, PPO=0x0D, CO=0, CI=0x05, CT=0x0D, Cache Hit?= Y, Byte =0x36
- 4. PA = 0x0254, PPN=0x14, PPO=0x0D, CO=0, CI=0x05, CT=0x0D, Cache Hit?= N, Byte =0x26

The correct answer is: PA = 0x0354, PPN=0x0D, PPO=0x14, CO=0, CI=0x05, CT=0x0D, Cache Hit?= Y, Byte =0x36

Question 175
Not answered
Marked out of 5.00

The file descriptor associated with standard output (stdout) is ....

Select one:
1. 1
2. 2
3. 0
4. 3

The correct answer is: 1

Question 176
Not answered
Marked out of 10.00

.... Returns the number of the signal that caused the child process to terminate.

Select one:
1. WIFSIGNALED(status)
2. WIFEXITED(status)
3. WEXITSTATUS(status)
4. WTERMSIG(status)

The correct answer is: WTERMSIG(status)

Question 177
Not answered
Marked out of 10.00

What happens when the following main program is executed clearly indicating when signal handler is called and provide the output produced.

```
int main()
{
        signal(SIGKILL, handler);
        pause();
        exit(0);
 }
void handler(int sig)
{
        printf("Caught SIGKILL\n");
        exit(0);
}
```

Select one:
1. SIGKILL caught, output: Caught SIGKILL
2. SIGKILL cannot be caught, output: Caught SIGKILL
3. SIGKILL cannot be caught, output: error
4. SIGKILL caught, output: error

The correct answer is: SIGKILL cannot be caught, output: error

Question 178
Not answered
Marked out of 5.00

The function to get the PID of the parent process of a calling process is ....

Select one:
1. getppid
2. getpid
3. getpgrp
4. getppgrp

The correct answer is: getppid

Question 179
Not answered
Marked out of 10.00

In which one of the Classical Inter-Process Communication(IPC) problems, three important issues such as deadlock, saturation and low concurrency are to be tackled ....

Select one:
1. Readers-Writers Problem
2. Dining Philosophers Problem
3. Bounded Buffer Problem
4. Producer-Consumer Problem

The correct answer is: Dining Philosophers Problem

Question 180
Not answered
Marked out of 10.00

Online airline reservation system is an example of ....

Select one:
1. Producer-Consumer Problem
2. Dining Philosophers Problem
3. Travelling Salesman Problem
4. Readers-Writers Problem

The correct answer is: Readers-Writers Problem

Question 181
Not answered
Marked out of 10.00

Semaphores are a fundamental mechanism for enforcing ....

Answer:

The correct answer is: mutual exclusion

Question 182

Not answered

Marked out of 20.00

For the following program given, Draw the process graph and Write the statements executed by only the parent process.

```
int main()
{
    pid_t pid;
    int x = 1;
    pid = Fork();
    if (pid == 0) {
        x=x+100;
        printf("child : x=%d\n", --x);
        exit(0);
    }
    x=x-100 ;
    printf("parent: x=%d\n", ++x);
    exit(0);
}
```

Select one:

1.

```
pid = Fork();
x=x+100;
printf("child : x=%d\n", --x);
x=x-100 ;
printf("parent: x=%d\n", ++x);
exit(0);
```

2.

```
pid = Fork();
x=x+100 ;
printf("parent: x=%d\n", ++x);
exit(0);
```

3.

```
pid = Fork();
x=x+100;
printf("child : x=%d\n", --x);
exit(0);
```

4.

```
pid = Fork();
x=x-100 ;
printf("parent: x=%d\n", ++x);
exit(0);
```

The correct answer is:

```
pid = Fork();
x=x-100 ;
printf("parent: x=%d\n", ++x);
exit(0);
```

Question 183
Not answered
Marked out of 10.00

Every peer thread in a process contains ……………………………………………….

Select one:
- 1. Data registers, code, data, stack
- 2. Data registers, code, data, heap, stack
- 3. Data registers, code, data, kernel context
- 4. Data registers, Condition Codes, SP, PC, stack

The correct answer is: Data registers, Condition Codes, SP, PC, stack

Question 184
Not answered
Marked out of 10.00

The notion of mapping a set of contiguous virtual pages to an arbitrary location in an arbitrary file is known as …………

Answer:

The correct answer is: memory mapping

Question 185
Not answered
Marked out of 5.00

………. functions are async-signal-safe and can be used safely in signal handlers

Answer:

The correct answer is: Unix I/O

Question 186
Not answered
Marked out of 5.00

………. exception returns control to "current" instruction after the execution of the corresponding exception handler.

Select one:
- 1. Trap
- 2. Abort
- 3. Any
- 4. Fault

The correct answer is: Fault

Question 187
Not answered
Marked out of 5.00

The file descriptor associated with standard input(stdin) is ....

Select one:
1. 0
2. 2
3. 3
4. 1

The correct answer is: 0

Question 188
Not answered
Marked out of 5.00

Identify the absolute pathname from the following:

Select one:
1. ../../usr/bin/vim
2. /usr/include/sys/unistd.h
3. ../../home/droh/hello.c
4. ../dev//tty1

The correct answers are: /usr/include/sys/unistd.h, ../../home/droh/hello.c

Question 189
Not answered
Marked out of 10.00

Function is ………………………….. if either reentrant (e.g., all variables stored on stack frame) or non-interruptible by signals.

Select one:
1. sync-signal-unsafe
2. sync-signal-safe
3. async-signal-unsafe
4. async-signal-safe

The correct answer is: async-signal-safe

Question 190
Not answered
Marked out of 10.00

Consider a virtual memory system having a 48-bit virtual address space partitioned into 16KB pages. Determine the number of bits in VPN and VPO

Select one:
1. VPN=40 bits VPO=8 bits
2. VPN=14 bits VPO=34 bits
3. VPN=34 bits VPO=14 bits
4. VPN=24 bits VPO=24 bits

The correct answer is: VPN=34 bits VPO=14 bits

Question 191
Not answered
Marked out of 10.00

Semaphore invariant creates a .... region that encloses unsafe region and that cannot be entered by any trajectory.

Answer:

The correct answer is: forbidden

Question 192
Not answered
Marked out of 5.00

The ASCII character corresponding to Line Feed (LF) is ....

Select one:
1. 0x0d
2. 0x0a
3. 0x0c
4. 0x0b

The correct answer is: 0x0a

Question 193
Not answered
Marked out of 10.00

Consider a virtual memory system having a 32-bit virtual address space partitioned into 4KB pages.

Determine the number of bits in VPN and VPO

Select one:
1. VPN=24 bits VPO=8 bits
2. VPN=12 bits VPO=20 bits
3. PN=16 bits VPO=16 bits
4. VPN=20 bits VPO=12 bits

The correct answer is: VPN=20 bits VPO=12 bits

Question 194
Not answered
Marked out of 10.00

Two signals which cannot be caught by a process are ....

Select one:
1. SIGSTP & SIGKILL
2. SIGSTP & SIGQUIT
3. SIGQUIT & SIGINT
4. SIGKILL & SIGCHLD

The correct answer is: SIGSTP & SIGKILL

Question 195
Not answered
Marked out of 5.00

……………………… occurs when greedy high priority processes make shared resources unavailable for long periods. The low-priority processes requiring these resources will have to wait indefinitely and sometimes never get a chance to be serviced.

Select one:
1. Race condition
2. Live lock
3. Starvation
4. Deadlock

The correct answer is: Starvation

Question 196
Not answered
Marked out of 30.00

Reorder the following steps in the correct order to indicate address translation performed by the CPU Hardware along with the operating system kernel when there is a page fault.

a) MMU fetches PTE from page table in memory

b) Processor sends virtual address to MMU

c) Page Fault Handler first identifies a victim page in memory to be replaced (and, if dirty, pages it out to disk)

d) Handler returns to original process, restarting faulting instruction

e) If Valid bit in PTE is zero, then MMU triggers page fault exception

f) Page Fault Handler brings the required new page from Hard Disk and loads into memory and updates PTE in memory Page Table.

Select one:
1. a),b),c),d),e),f)
2. b),a),e),c),f),d)
3. b),e),a),f),c),d)
4. a),b),c),e),f),d)

The correct answer is: b),a),e),c),f),d)

Question 197

Not answered

Marked out of 20.00

Consider the following program where main thread creates 5 threads.

Identify all the shared variables in the following program.

```
char **ptr1;
char **ptr2;
int  count;

int main()
{
   long i;
   pthread_t tid[5];

   char *msgs1[2] = {
      "Hello from thread0",
      "Hello from thread1"
   };
  char *msgs2[3] = {
      "Hello from thread3",
      "Hello from thread4",
       "Hello from thread5"
   };
   count=100;
   ptr1 = msgs1;
   ptr2 = msgs2;

   for (i = 0; i < 2; i++)
      pthread_create(&tid[i],  NULL, thread1, (void *)i);
  for (i = 2; i < 5; i++)
      pthread_create(&tid[i],  NULL, thread2, (void *)i);
   printf("THREAD MAIN EXITING\n");
   Pthread_exit(NULL);
}
```

```
void *thread1(void *vargp1)
{
long myid1 = (long)vargp1;
 static int cnt1 = 0;

count *= ++cnt1;
printf("[%ld]:  %s (cnt=%d) – count=%d \n",
        myid1, ptr1[myid1], cnt1, count);
 printf("THREAD %ld RETURNING \n", myid1);
 return NULL;
}

void *thread2(void *vargp2)
{
long myid2 = (long)vargp2;
static int cnt2 = 3;

count *= cnt2--;
printf("[%ld]:  %s (cnt=%d) – count=%d \n", myid2,
ptr2[myid2-2], cnt2, count);
printf("THREAD %ld RETURNING \n", myid2);
return NULL;
}
```

Select one:

1. ptr1, ptr2, msg1, msg2, count, tid
2. msg1, msg2, count, cnt1, cnt2
3. ptr1, ptr2, msg1, msg2, count, cnt1, cnt2
4. ptr1, ptr2, msg1, msg2, count

The correct answer is: ptr1, ptr2, msg1, msg2, count

Question 198

Not answered

Marked out of 5.00

....uses main memory efficiently by treating it as a cache for an address space stored on disk, keeping only the active areas in main memory, and transferring data back and forth between disk and memory as needed.

Select one:

1. Direct Mapped Cache
2. Fully Associative Cache
3. Set Associative Cache
4. Virtual Memory

The correct answer is: Virtual Memory

Question 199
Not answered
Marked out of 5.00

................., a pioneer of concurrent programming, proposed a classic solution to the problem of synchronizing different execution threads based on a special type of variable called a semaphore.

Select one:
1. Dennis Ritchie
2. Edsger Dijkstra
3. Richard Stallman
4. Ken Thompson

The correct answer is: Edsger Dijkstra

Question 200
Not answered
Marked out of 5.00

The Page table which is used by MMU to translate virtual address to physical address is stored in ……………………………..

Select one:
1. Split Cache L1
2. memory management unit (MMU)
3. Unified Cache L2
4. Main Memory

The correct answer is: Main Memory

Question 201
Not answered
Marked out of 5.00

The ID number of x86-64 System call exit is ………

Select one:
1. 62
2. 60
3. 57
4. 59

The correct answer is: 60

Question 202
Not answered
Marked out of 5.00

Applications that use application-level concurrency are known as …………… programs.

Select one:
1. Concurrent
2. Atomic
3. Modular
4. Sequential

The correct answer is: Concurrent

Question 203
Not answered
Marked out of 10.00

Function exit is …………………………………….

Select one:

1. called once, returns one or more times
2. called once, returns only once
3. called once, returns twice
4. called once, never returns

The correct answer is: called once, never returns

Question 204
Not answered
Marked out of 20.00

Given a Virtual Memory System having 64-bit virtual address, 4KB Page size and using a single Page table with 8 byte PTE (size of each Page table entry), determine the amount of Main Memory required to store a single Page Table.

Select one:

1. 32PB
2. 16GB
3. 32GB
4. 16PB

The correct answer is: 32PB

Question 205
Not answered
Marked out of 5.00

……………… is considered mainly as a mechanism that the operating system kernel uses to run multiple application programs.

Select one:

1. fork
2. Exception
3. Interrupt
4. Concurrency

The correct answer is: Concurrency

Question 206
Not answered
Marked out of 15.00

What Signal is sent to a process by the kernel when a user presses the following key combinations:

a)	**Ctrl c : ....**
b)	**Ctrl z : ....**
c)	**Ctrl \ : ....**

Select one:

1. a) SIGINT b) SIGQUIT c) SIGSTP
2. a) SIGSTP b) SIGINT c) SIGQUIT
3. a) SIGQUIT b) SIGSTP c) SIGINT
4. a) SIGINT b) SIGSTP c) SIGQUIT

The correct answer is: a) SIGINT b) SIGSTP c) SIGQUIT

Question 207
Not answered
Marked out of 10.00

The operation V(s) is defined as ....

Select one:

1. If s==1 then
    s=s -1
    Enter Critical region/section
   Else
    Wait or block
2. s = s - 1
3. s = s + 1
4. If s==1 then
    Wait or block
   Else
    s=s -1
    Enter Critical region/section

The correct answer is: s = s + 1

Question 208
Not answered
Marked out of 5.00

A .... thread can be reaped and killed by other threads.

Answer:

The correct answer is: joinable

Question 209

Not answered

Marked out of 15.00

Consider the following program where main thread creates 2 threads.

Identify all the shared variables in the following program.

```
char **ptr;  /* global var */
int main()
{
    long i;
    pthread_t tid[2];
    char *msgs[2] = {
        "Hello from foo",
        "Hello from bar"
    };
    ptr = msgs;
    for (i = 0; i < 2; i++)
        pthread_create(&tid[I], NULL, thread,
(void *)i);
    pthread_exit(NULL);
}

void *thread(void *vargp)
{
    long myid = (long)vargp;
    static int cnt = 0;

    printf("[%ld]:  %s (cnt=%d)\n",
         myid, ptr[myid], ++cnt);
    return NULL;
}
```

Select one:

1. i, cnt, myid

2. ptr, cnt, msgs

3. i, tid, myid

4. i, cnt, msgs

The correct answer is: ptr, cnt, msgs

Question 210

Not answered

Marked out of 15.00

Consider the following program where main thread creates 5 threads.

Identify all the global variables in the following program.

```
char **ptr1;
char **ptr2;
int  count;

int main()
{
   long i;
   pthread_t tid[5];

   char *msgs1[2] = {
      "Hello from thread0",
      "Hello from thread1"
   };
  char *msgs2[3] = {
      "Hello from thread3",
      "Hello from thread4",
       "Hello from thread5"
   };
   count=100;
   ptr1 = msgs1;
   ptr2 = msgs2;

   for (i = 0; i < 2; i++)
      pthread_create(&tid[i],  NULL, thread1, (void *)i);
  for (i = 2; i < 5; i++)
      pthread_create(&tid[i],  NULL, thread2, (void *)i);
   printf(“THREAD MAIN EXITING\n”);
   Pthread_exit(NULL);
}
```

```
void *thread1(void *vargp1)
{
long myid1 = (long)vargp1;
 static int cnt1 = 0;

count *= ++cnt1;
printf("[%ld]:  %s (cnt=%d) – count=%d \n",
        myid1, ptr1[myid1], cnt1, count);
 printf(“THREAD %ld RETURNING \n”, myid1);
 return NULL;
}

void *thread2(void *vargp2)
{
long myid2 = (long)vargp2;
static int cnt2 = 3;

count *= cnt2--;
printf("[%ld]:  %s (cnt=%d) – count=%d \n",  myid2,
ptr2[myid2-2], cnt2, count);
printf(“THREAD %ld RETURNING \n”, myid2);
return NULL;
}
```

Select one:

1. ptr1, ptr2, cnt1, cnt2, count
2. msg1, msg2, count
3. ptr1, ptr2, msg1, msg2, cnt1, cnt2, count
4. ptr1, ptr2, count

The correct answer is: ptr1, ptr2, count

Question 211

Not answered

Marked out of 10.00

Given a Virtual Memory System having 64-bit virtual address, 4KB Page size and using a single Page table, determine the number of bits in VPN and VPO fields of the virtual address.

Select one:

1. VPN=48 bits VPO= 16 bits
2. VPN=32 bits VPO= 32 bits
3. VPN=52 bits VPO= 12 bits
4. VPN=12 bits VPO= 52 bits

The correct answer is: VPN=52 bits VPO= 12 bits

Question 212
Not answered
Marked out of 10.00

What happens when the following main program is executed clearly indicating when signal handler is called and provide the output produced.

```
int main()
{
        signal(SIGSTP, handler);
        pause();
        exit(0);
}
void handler(int sig)
{
        printf("Caught SIGSTP\n");
        exit(0);
}
```

Select one:

1. SIGSTP caught, output: Caught SIGSTP
2. SIGSTP cannot be caught, output: error
3. SIGSTP cannot be caught, output: Caught SIGSTP
4. SIGSTP caught, output: error

The correct answer is: SIGSTP cannot be caught, output: error

Question 213
Not answered
Marked out of 5.00

A .... thread cannot be reaped or killed by other threads.

Answer:

The correct answer is: detached

Question 214
Not answered
Marked out of 5.00

The POSIX library function to create a peer thread is ....

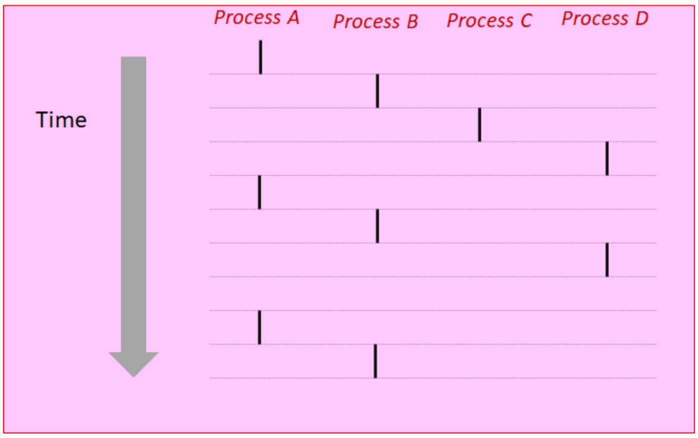
Select one:

1. pthread_self
2. pthread_create
3. pthread_join
4. pthread_detach

The correct answer is: pthread_create

Question 215
Not answered
Marked out of 15.00

Given the following timing diagram showing the execution flows for four different processes A, B, C & D. Identify the processes which are running concurrently and the processes running sequentially.



Select one:

1. Concurrent A & B, A & C, A & D, Sequential B & C, C & D
2. Concurrent: A & B, A & C, A & D, B & C, B & D, Sequential: C & D
3. Concurrent A & B, A & D, Sequential B & C, C & D
4. Concurrent A & B, Sequential B & C, C & D

The correct answer is: Concurrent: A & B, A & C, A & D, B & C, B & D, Sequential: C & D

Question 216
Not answered
Marked out of 5.00

Example of an asynchronous interrupt is ....

Select one:

1. divide by zero interrupt
2. keyboard interrupt
3. overflow interrupt
4. breakpoint interrupt

The correct answer is: keyboard interrupt

Question 217
Not answered
Marked out of 15.00

Given a Virtual Memory System having 64-bit virtual address, 4KB Page size and using a single Page table,

Determine the number of entries in the page table (i.e., number of PTEs).

Select one:

1. PTEs = 4Giga entries
2. PTEs = 8Giga entries
3. PTEs = 4Peta entries
4. PTEs = 8Peta entries

The correct answer is: PTEs = 4Peta entries

Question 218

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one infeasible output

```
int main()
{
        printf("ONLINE EDUCATION");
         if (fork() == 0 )
            printf("SYSTEMS PROGRAMMING");
         if (fork() != 0)
              printf("GOOD LUCK");
               printf("BYE \n");
}
```

Select one:

1. ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
GOOD LUCK
BYE
BYE
BYE

2. ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
GOOD LUCK
BYE
BYE
BYE

3. ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
BYE
GOOD LUCK
BYE
BYE

4. ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
BYE
BYE
BYE
GOOD LUCK

The correct answer is: ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
BYE
BYE
BYE
GOOD LUCK

Question 219

Not answered

Marked out of 10.00

Consider a virtual memory system having a 32-bit virtual address space partitioned into 8KB pages.

Determine the number of bits in VPN and VPO

Select one:

1. VPN=13 bits   VPO=19 bits
2. VPN=16 bits   VPO=16 bits
3. VPN=24 bits   VPO=8 bits
4. VPN=19 bits   VPO=13 bits

The correct answer is: VPN=19 bits   VPO=13 bits

Question 220

Not answered

Marked out of 25.00

Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Write the assembly language instructions in the critical region .

```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

1. L: movq cnt(%rip),%rdx
   U: addq $1, %rdx
   S: movq %rdx, cnt(%rip)
2. L: movq %rdx, cnt(%rip)
   U: addq $1, %rax
   S: addq $1, %rdx
   movq %rdx, cnt(%rip)
3. L: addq $1, %rdx
   U: movq cnt(%rip), %rdx
   S: addq $1, %rax
   movq %rdx, cnt(%rip)
4. L: movq %rdx, cnt(%rip)
   U: addq $1, %rdx
   S: movq cnt(%rip),%rdx

The correct answer is: L: movq cnt(%rip),%rdx
U: addq $1, %rdx
S: movq %rdx, cnt(%rip)

Question 221
Not answered
Marked out of 5.00

Logical control flows are .... if they overlap in time.

Answer:

The correct answer is: concurrent

Question 222
Not answered
Marked out of 5.00

Identify Class of thread-safe functions: ....

Select one:
1. Functions that keep state across multiple invocations
2. Functions that do not return a pointer to a static variable
3. Functions that do not protect shared variables
4. Functions that call thread-unsafe functions

The correct answer is: Functions that do not return a pointer to a static variable

Question 223
Not answered
Marked out of 5.00

Synchronous exception Aborts are ....

Select one:
1. Intentional & Unrecoverable
2. Unintentional & Unrecoverable
3. Intentional
4. Unintentional but possibly recoverable

The correct answer is: Unintentional & Unrecoverable

Question 224
Not answered
Marked out of 5.00

An example of Synchronous exception Abort is ....

Select one:
1. System Call
2. Page fault
3. Illegal Instruction
4. Floating point exception

The correct answer is: Illegal Instruction

Question 225

Not answered

Marked out of 30.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

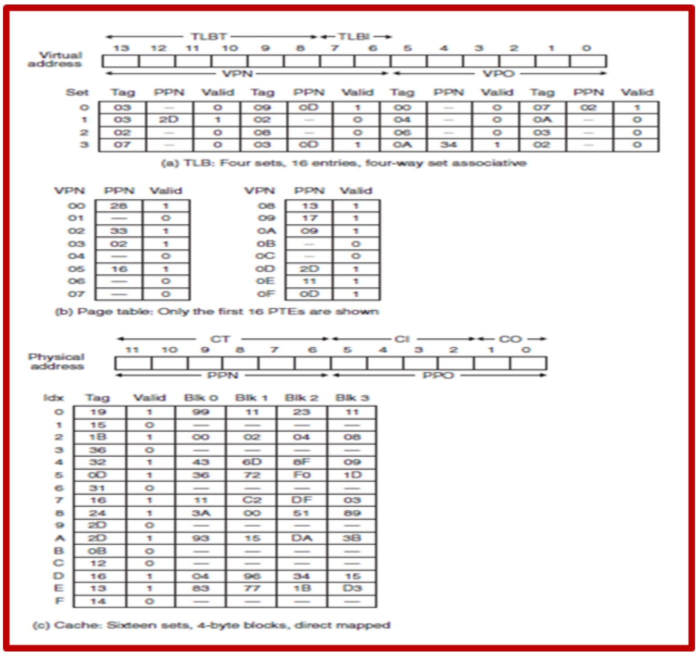
Cache block is byte addressable.

Processor generates a virtual address VA = 0x0320, Translate this to physical address PA using TLB or Page Table

Determine the values (in hex) in the following fields:

Physical Address PA = ..., PPN = ..., PPO= .....

CO=....., CI=......, CT=......, Cache Hit?.... (Y/N), Byte (read from cache)=......



Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	—	0	09	0D	1	00	—	0	07	02	1
1	03	2D	1	02	—	0	04	—	0	0A	—	0
2	02	—	0	08	—	0	06	—	0	03	—	0
3	07	—	0	03	0D	1	0A	34	1	02	—	0

(a) TLB: Four sets, 16 entries, four-way set associative

VPN	PPN	Valid	VPN	PPN	Valid
00	28	1	08	13	1
01	—	0	09	17	1
02	33	1	0A	09	1
03	02	1	0B	—	0
04	—	0	0C	—	0
05	16	1	0D	2D	1
06	—	0	0E	11	1
07	—	0	0F	0D	1

(b) Page table: Only the first 16 PTEs are shown

Idx	Tag	Valid	Blk 0	Blk 1	Blk 2	Blk 3
0	19	1	99	11	23	11
1	15	0	—	—	—	—
2	1B	1	00	02	04	08
3	36	0	—	—	—	—
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	—	—	—	—
7	16	1	11	C2	DF	03
8	24	1	3A	00	51	89
9	2D	0	—	—	—	—
A	2D	1	93	15	DA	3B
B	0B	0	—	—	—	—
C	12	0	—	—	—	—
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	—	—	—	—

(c) Cache: Sixteen sets, 4-byte blocks, direct mapped

Select one:

1. PA = 0x0A20, PPN=0x20, PPO=0x28, CO=0, CI=0x08, CT=........, Cache Hit?= N, Byte = ...(Disk)
2. PA = ..., PPN=........, PPO=0x20, CO=0, CI=0x08, CT=........, Cache Hit?= N, Byte = ...(Disk)
3. PA = 0x0A20, PPN=0x28, PPO=0x20, CO=0, CI=0x08, CT=........, Cache Hit?= N, Byte = ...(Disk)
4. PA = ..., PPN=........, PPO=0x0, CO=0x20, CI=0x08, CT=........, Cache Hit?= N, Byte = ...(Disk)

The correct answer is: PA = ..., PPN=........, PPO=0x20, CO=0, CI=0x08, CT=........, Cache Hit?= N, Byte = ...(Disk)

Question 226
Not answered
Marked out of 10.00

……….. are considered as a hybrid of the other two approaches, scheduled by the kernel like process flows, and sharing the same virtual address space like I/O multiplexing flows.

Answer:

The correct answer is: Threads

Question 227
Not answered
Marked out of 5.00

When a process creates a new file by calling the open function with some mode argument,

then the access permission bits of the file are set to ……….

Select one:
1. mode & umask
2. mode & ~umask
3. ~mode & umask
4. ~mode & ~umask

The correct answer is: mode & ~umask

Question 228
Not answered
Marked out of 5.00

Programs free allocated heap blocks by calling the ……………… function.

Select one:
1. valloc
2. palloc
3. free
4. malloc

The correct answer is: free

Question 229
Not answered
Marked out of 10.00

If the calling process has no children, the waitpid returns ……… and set errno to……

Select one:
1. 0, ECHILD
2. -1, ECHILD
3. -1, EINTR
4. 0, EINTR

The correct answer is: -1, ECHILD

Question 230
Not answered
Marked out of 10.00

Indicate the classical problem encountered in the following typical Scenario :

Processes 1 and 2 need two resources (A and B) to proceed

Process 1 acquires A, waits for B

Process 2 acquires B, waits for A

Both will wait forever!

Select one:
- 1. Deadlock
- 2. Race Condition
- 3. Livelock
- 4. Starvation

The correct answer is: Deadlock

Question 231
Not answered
Marked out of 10.00

A fork system call returns ………… to the child process and ………………………………..to the parent process.

Select one:
- 1. 0, PID of child
- 2. PID of child, 0
- 3. PID of parent, 0
- 4. PID of parent, 1

The correct answer is: 0, PID of child

Question 232
Not answered
Marked out of 10.00

You are given a process group with pgid = 2000 having a parent process with pid = 2000 and two child processes,

child1 with pid=2001 & child2 with pid=2002 which are running at the foreground.

Write a command to Kill the process group.

Select one:
- 1. kill -9 2002
- 2. kill -9 2000
- 3. kill -9 -2000
- 4. kill -9 2001

The correct answer is: kill -9 -2000

Question 233
Not answered
Marked out of 5.00

One similarity between thread and process ....

Select one:
- 1. Both thread & process are context switched
- 2. Both thread & process do not share code and data
- 3. Both thread & process are less expensive.
- 4. Both thread & process are not context switched

The correct answer is: Both thread & process are context switched

Question 234
Not answered
Marked out of 10.00

.... is still possible in First Readers-Writers problem implementation

Select one:
- 1. Deadlock
- 2. Race Condition
- 3. Livelock
- 4. Starvation

The correct answer is: Starvation

Question 235
Not answered
Marked out of 5.00

The C standard library (libc.so) contains a collection of higher-level .... functions.

Answer:

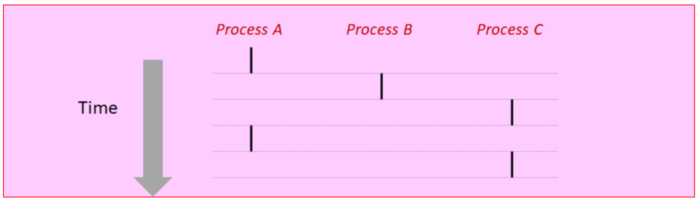
The correct answer is: standard I/O

Question 236

Not answered

Marked out of 10.00

Given the following timing diagram showing the execution flows for three different processes A, B, & C. Identify the processes which are running concurrently and the processes running sequentially.



Select one:

1. Concurrent A & B, A & C, Sequential B & C, A & C
2. Concurrent A & B, A & C,   Sequential B & C
3. Concurrent A & B, Sequential B & C, A & C
4. Concurrent A & B, Sequential A & C, B & C

The correct answer is: Concurrent A & B, A & C,   Sequential B & C

Question 237

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one feasible output

```
int main()
{
          printf("ONLINE EDUCATION");
           if (fork() == 0 )
              printf("SYSTEMS PROGRAMMING");
           if (fork() != 0)
                printf("GOOD LUCK");
                 printf("BYE \n");
}
```

Select one:

1. ONLINE EDUCATION
BYE
GOOD LUCK
BYE
SYSTEMS PROGRAMMING
BYE
BYE
GOOD LUCK

2. ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
BYE
BYE
BYE
GOOD LUCK

3. ONLINE EDUCATION
GOOD LUCK
SYSTEMS PROGRAMMING
BYE
BYE
BYE
BYE
GOOD LUCK

4. ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
GOOD LUCK
BYE
BYE
BYE

The correct answer is: ONLINE EDUCATION
SYSTEMS PROGRAMMING
GOOD LUCK
BYE
GOOD LUCK
BYE
BYE
BYE

Question 238
Not answered
Marked out of 5.00

The ID number of x86-64 System call fork is ·········

Select one:
- 1. 60
- 2. 62
- 3. 59
- 4. 57

The correct answer is: 57

Question 239
Not answered
Marked out of 5.00

.... Returns true if the child terminated normally, via a call to exit or a return

Select one:
- 1. WEXITSTATUS(status)
- 2. WIFEXITED(status)
- 3. WIFSIGNALED(status)
- 4. WTERMSIG(status)

The correct answer is: WIFEXITED(status)

Question 240
Not answered
Marked out of 5.00

.... files are regular files with only ASCII or Unicode characters.

Answer:

The correct answer is: Text

Question 241
Not answered
Marked out of 5.00

Programs allocate blocks from the heap by calling the ·················· function.

Select one:
- 1. valloc
- 2. palloc
- 3. malloc
- 4. free

The correct answer is: malloc

Question 242

Not answered

Marked out of 10.00

Function setjmp is ……………………………………….

Select one:

1. called once, never returns
2. called once, returns twice
3. called once, returns one or more times
4. called once, returns only once

The correct answer is: called once, returns one or more times

Question 243
Not answered
Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one infeasible output.

```
int main()
{
    printf("L0\n");
    fork();
    printf("L1\n");
    fork();
    printf("Bye\n");
}
```

Select one:

1. L0
L1
Bye
L1
Bye
Bye
Bye

2. L0
Bye
L1
Bye
L1
Bye
Bye

3. L0
L1
L1
Bye
Bye
Bye
Bye

4. L0
L1
Bye
Bye
L1
Bye
Bye

The correct answer is: L0
Bye
L1
Bye
L1
Bye
Bye

Question 244
Not answered
Marked out of 10.00

Consider a virtual memory system having a 32-bit virtual address space partitioned into 4MB pages.

Determine the number of bits in VPN and VPO

Select one:
1. VPN=10 bits   VPO=22 bits
2. VPN=22 bits   VPO=10 bits
3. VPN=24 bits   VPO=8 bits
4. VPN=16 bits   VPO=16 bits

The correct answer is: VPN=10 bits   VPO=22 bits

Question 245
Not answered
Marked out of 10.00

The operation P(s) is defined as ....

Select one:

1. If s==0 then
    s=s -1
    Enter Critical region/section
 Else
    Wait or block
2. If s==1 then
    s=s -1
    Enter Critical region/section
 Else
    Wait or block
3. s = s + 1
4.

 If s==1 then
    Wait or block
 Else
    s=s -1
    Enter Critical region/section

The correct answer is: If s==1 then
  s=s -1
  Enter Critical region/section
Else
  Wait or block

Question 246
Not answered
Marked out of 10.00

The .... function suspends a process for a specified period of time.

Answer:

The correct answer is: sleep

Question 247

Not answered

Marked out of 15.00

Identify three most important examples of Classical IPC Problems ....

Select one or more:

- 1. Producer-Consumer Problem
- 2. Dining Philosophers Problem
- 3. Readers-Writers Problem
- 4. Travelling Salesman Problem

The correct answers are: Producer-Consumer Problem, Readers-Writers Problem, Dining Philosophers Problem

Question 248

Not answered

Marked out of 10.00

A .... depicts the discrete execution state space of concurrent threads.

Answer:

The correct answer is: progress graph

Question 249

Not answered

Marked out of 20.00

Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Rewrite the thread function using semaphores P(&mutex) and V(&mutex) to control access to cnt shared variable to provide proper synchronization.

```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

1.
```
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

      P(&mutex);
      for (i = 0; i < niters; i++)
         cnt++;
      V(&mutex);
   return NULL;
}
```

2.
```
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

      for (i = 0; i < niters; i++)
      {

       V(&mutex);
        cnt++;
       P(&mutex);
      }
      return NULL;
}
```

3.
```
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

      for (i = 0; i < niters; i++)
      {
```

```
        P(&mutex);
        cnt++;
        V(&mutex);
        }

        return NULL;
    }
```

4. void *thread(void *vargp)

```
    {
      long i, niters =    *((long *)vargp);

        P(&mutex);
        for (i = 0; i < niters; i++)
        V(&mutex);
        cnt++;
      return NULL;
    }
```

The correct answer is: void *thread(void *vargp)

```
{
    long i, niters =    *((long *)vargp);

        for (i = 0; i < niters; i++)
        {
         P(&mutex);
         cnt++;
         V(&mutex);
         }

        return NULL;
}
```

Question 250

Not answered

Marked out of 5.00

…………………………….. function is used to move the file position indicator to the BEGINNING of the file.

Select one:

1. lseek(fd, 0L, 0);
2. lseek(fd, 0L, -1);
3. lseek(fd, 0L, 1);
4. lseek(fd, 0L, 2);

The correct answer is: lseek(fd, 0L, 0);

Question 251

Not answered

Marked out of 10.00

Short counts never occur in these situations:

Select one or more:

1. Reading and writing network sockets
2. Encountering (end-of-file) EOF on reads
3. Writing to disk files
4. Reading from disk files (except for EOF)
5. Reading text lines from a terminal

The correct answers are: Reading from disk files (except for EOF), Writing to disk files

Question 252
Not answered
Marked out of 5.00

.... exception returns control to “next” instruction after the execution of the corresponding exception handler.

Select one:
1. Abort
2. Trap
3. Any
4. Fault

The correct answer is: Trap

Question 253
Not answered
Marked out of 5.00

The POSIX library function to initialize a global variable or state in a thread ....

Select one:
1. pthread_cancel
2. pthread_once
3. pthread_join
4. pthread_exit

The correct answer is: pthread_once

Question 254
Not answered
Marked out of 5.00

An example of Synchronous exception Trap is ....

Select one:
1. System Call
2. Page fault
3. Illegal Instruction
4. Floating point exception

The correct answer is: System Call

Question 255
Not answered
Marked out of 5.00

Example of a synchronous interrupt is ....

Select one:
1. interrupt on NMI pin
2. divide by zero interrupt
3. keyboard interrupt
4. timer interrupt

The correct answer is: divide by zero interrupt

Question 256
Not answered
Marked out of 10.00

Consider a virtual memory system having a 48-bit virtual address space partitioned into 8KB pages.

Determine the number of bits in VPN and VPO

Select one:

1. VPN=35 bits   VPO=13 bits
2. VPN=40 bits   VPO=8 bits
3. VPN=24 bits   VPO=24 bits
4. VPN=13 bits   VPO=35 bits

The correct answer is: VPN=35 bits   VPO=13 bits

Question 257
Not answered
Marked out of 10.00

Indicate what happens when the following instructions are executed :

#define DEF_MODE   S_IRUSR| S_IWUSR| S_IXUSR| S_IRGRP| S_IWGRP| S_IROTH| S_IWOTH

#define DEF_UMASK   S_IWGRP|S_IWOTH | S_IROTH

umask(DEF_UMASK);

fd1 = Open("list.txt", O_CREAT|O_TRUNC|O_RDWR, DEF_MODE);

Select one:

1. create a file list.txt for both reading & writing with read & write access for both group and others
2. create a file list.txt for both reading & writing with only read access for group and no access for others
3. create a file list.txt for both reading & writing with write access for group and read access for others
4. create a file list.txt for both reading & writing with no access for both group and others

The correct answer is: create a file list.txt for both reading & writing with only read access for group and no access for others

Question 258
Not answered
Marked out of 5.00

Fatal Hardware Error raises abort exception and the kernel sends ....signal to user process and the User process exits with “segmentation fault”

Answer: ✗

The correct answer is: SIGSEGV

Question 259
Not answered
Marked out of 5.00

.... functions are not async-signal-safe, and not appropriate for signal handlers

Answer:

The correct answer is: Standard I/O

Question 260
Not answered
Marked out of 5.00

Dedicated hardware on the CPU chip called the .... translates virtual addresses on the fly, using a look-up table stored in main memory whose contents are managed by the operating system

Select one:

1. Split Cache L1
2. Unified Cache L3
3. memory management unit (MMU)
4. Unified Cache L2

The correct answer is: memory management unit (MMU)

Question 261

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one feasible output

```
Int main()
{
 printf("L0\n");
   if (fork() != 0) {
       printf("L1\n");
       if (fork() != 0) {
           printf("L2\n");
               }
   }
   printf("Bye\n");
}
```

Select one:

1. L0
Bye
Bye
L1
Bye
L2

2. L0
Bye
Bye
Bye
L1
L2

3. L0
L1
Bye
Bye
L2
Bye

4. L0
Bye
L1
Bye
Bye
L2

The correct answer is: L0
L1
Bye
Bye
L2
Bye

Question 262
Not answered
Marked out of 5.00

.. function is used by the parent to reap a child

Select one:

1. wstopped
2. wnohang
3. waitpid
4. wuntraced

The correct answer is: waitpid

Question 263
Not answered
Marked out of 5.00

The POSIX library function to perform P operation on a semaphore is ....

Select one:

1. sem_init
2. sem_wait
3. sem_assign
4. sem_post

The correct answer is: sem_wait

Question 264
Not answered
Marked out of 10.00

A program is .... if, for each pair of mutexes (s, t) in the program, each thread that holds both s and t simultaneously locks them in the same order.

Answer: ✗

The correct answer is: deadlock-free

Question 265

Not answered

Marked out of 20.00

In the First Readers-Writers Problem implementation,

(i) semaphore variable mutex is used to protect the global variable .... and

(ii) semaphore variable .... is used to protect access to the critical section.

Readers:

```
int readcnt;    /* Initially = 0 */
sem_t mutex, w; /* Initially = 1 */

void reader(void)
{
   while (1)
   {
              P(&mutex);
              readcnt++;
              if (readcnt == 1) /* First in */
              P(&w);
              V(&mutex);

              /* Critical section  */
              /* Reading happens  */

              P(&mutex);
              readcnt--;
              if (readcnt == 0) /* Last out */
               V(&w);
              V(&mutex);
   }
}
```

Writers:

```
void writer(void)
{
  while (1)
  {
    P(&w);

    /* Critical section */
    /* Writing happens  */

    V(&w);
  }
}
```

Select one:

1. (i) mutex, ii) w
2. (i) readcnt, ii) w
3. (i) w, ii) readcnt
4. (i) readcnt, ii) mutex

The correct answer is: (i) readcnt, ii) w

Question 266
Not answered
Marked out of 10.00

Given a Virtual Memory System having 64-bit virtual address, 2MB Page size and using a single Page table, determine the number of bits in VPN and VPO fields of the virtual address.

Select one:
1. VPN=56 bits, VPO= 8 bits
2. VPN=32 bits, VPO= 32 bits
3. VPN=43 bits, VPO= 21 bits
4. VPN=21 bits, VPO= 43 bits

The correct answer is: VPN=43 bits, VPO= 21 bits

Question 267
Not answered
Marked out of 5.00

……………………… occurs when two or more processes continually repeat the same interaction in response to changes in the other processes without doing any useful work. These processes are not in the waiting state, and they are running concurrently without progressing forward.

Select one:
1. Race condition
2. Live lock
3. Deadlock
4. Starvation

The correct answer is: Live lock

Question 268

Not answered

Marked out of 30.00

Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Write the corresponding assembly code for the counter loop in thread i with register mapping : %rdi = &niters, %rax=i, %rdx=cnt, %rcx=niters

```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =    *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

1.
```
    movq  (%rdi), %rcx
    testq %rcx,%rcx
    jle   .L2
    movq  $0, %rax
.L3:
    movq  %rdx, cnt(%rip)
    addq  $1, %rax
    addq  $1, %rdx
    movq  cnt(%rip),%rdx
    cmpq  %rcx, %rax
    jne   .L3
.L2:
```

2.
```
    movq  (%rdi), %rcx
    testq %rcx,%rcx
    jle   .L2
    movq  $0, %rax
.L3:
    movq  %rdx, cnt(%rip)
    addq  $1, %rdx
    movq  cnt(%rip),%rdx
    addq  $1, %rax
    cmpq  %rcx, %rax
    jne   .L3
.L2:
```

3.
```
    movq  (%rdi), %rcx
    testq %rcx,%rcx
    jle   .L2
    movq  $0, %rax
.L3:
    movq  cnt(%rip),%rdx
    addq  $1, %rdx
```

```
    movq  %rdx, cnt(%rip)
    addq  $1, %rax
    cmpq  %rcx, %rax
    jne   .L3
  .L2:
```

4.
```
    movq  %rcx, (%rdi)
    testq %rcx,%rcx
    jle   .L2
    movq  $0, %rax
  .L3:
    movq  %rdx,cnt(%rip)
    addq  $1, %rdx
    movq  cnt(%rip), %rdx
    addq  $1, %rax
    cmpq  %rcx, %rax
    jne   .L3
  .L2:
```

The correct answer is:

```
    movq  (%rdi), %rcx
    testq %rcx,%rcx
    jle   .L2
    movq  $0, %rax
.L3:
    movq  cnt(%rip),%rdx
    addq  $1, %rdx
    movq  %rdx, cnt(%rip)
    addq  $1, %rax
    cmpq  %rcx, %rax
    jne   .L3
.L2:
```

Question 269
Not answered
Marked out of 5.00

The default action taken by the kernel after a process receives the signal SIGCHLD is………………..

Select one:
1. Terminate with Dump
2. Terminate
3. Suspend
4. Ignore

The correct answer is: Ignore

Question 270
Not answered
Marked out of 10.00

Identify all the high level languages the use implicit allocator..............

Select one:
1. Java
2. Lisp
3. ML
4. All of the above

The correct answer is: All of the above

Question 271
Not answered
Marked out of 10.00

Given a Virtual Memory System having 48-bit virtual address, 64KB Page size and using a single Page table, determine the number of bits in VPN and VPO fields of the virtual address.

Select one:
1. VPN=40 bits VPO= 8 bits
2. VPN=16 bits VPO= 32 bits
3. VPN=24 bits VPO= 24 bits
4. VPN=32 bits VPO= 16 bits

The correct answer is: VPN=32 bits VPO= 16 bits

Question 272
Not answered
Marked out of 10.00

A .... models instruction execution as a transition from one state to another.

Answer:

The correct answer is: progress graph

Question 273
Not answered
Marked out of 5.00

Physical address is divided into two parts ...., ...

Select one:
1. PPN, PPO
2. VPA, PPA
3. VPN, VPO
4. LPA, RPA

The correct answer is: PPN, PPO

Question 274

Not answered

Marked out of 30.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 14 bits wide.

Physical addresses(PA) are 12 bits wide.

The page size is 64 bytes.

The TLB is four-way set associative with 16 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 4-byte line(block)size and 16 total blocks.

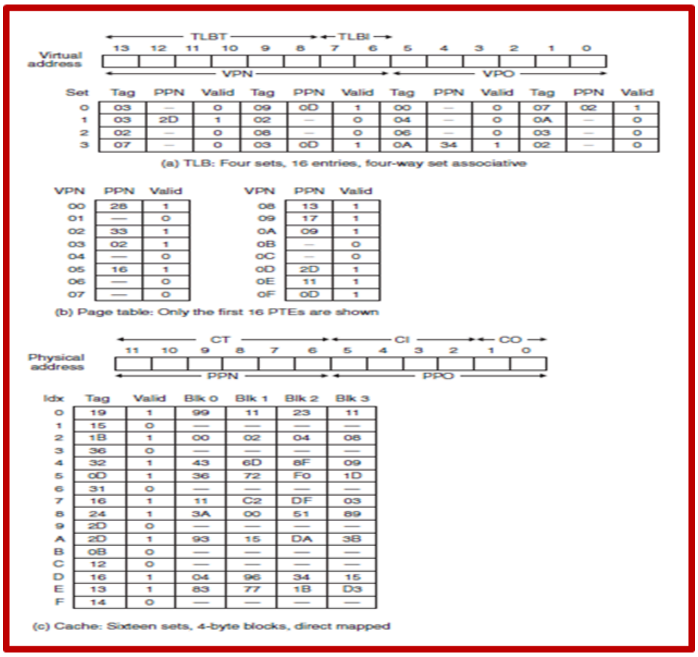
Cache block is byte addressable.

Processor generates a virtual address VA = 0x03D4, Translate this to physical address PA using TLB or Page Table

Determine the values (in hex) in the following fields:

VPN =...., VPO=....., TLBI=....., TLBT=....., TLB Hit?=....(Y/N), Page Fault?=....(Y/N), PPN=....,

Page Table access Required PTAR:..... (Y/N)



Set	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid	Tag	PPN	Valid
0	03	—	0	09	0D	1	00	—	0	07	02	1
1	03	2D	1	02	—	0	04	—	0	0A	—	0
2	02	—	0	08	—	0	06	—	0	03	—	0
3	07	—	0	03	0D	1	0A	34	1	02	—	0

(a) TLB: Four sets, 16 entries, four-way set associative

VPN	PPN	Valid	VPN	PPN	Valid
00	28	1	08	13	1
01	—	0	09	17	1
02	33	1	0A	09	1
03	02	1	0B	—	0
04	—	0	0C	—	0
05	16	1	0D	2D	1
06	—	0	0E	11	1
07	—	0	0F	0D	1

(b) Page table: Only the first 16 PTEs are shown

Idx	Tag	Valid	Blk 0	Blk 1	Blk 2	Blk 3
0	19	1	99	11	23	11
1	15	0	—	—	—	—
2	1B	1	00	02	04	08
3	36	0	—	—	—	—
4	32	1	43	6D	8F	09
5	0D	1	36	72	F0	1D
6	31	0	—	—	—	—
7	16	1	11	C2	DF	03
8	24	1	3A	00	51	89
9	2D	0	—	—	—	—
A	2D	1	93	15	DA	3B
B	0B	0	—	—	—	—
C	12	0	—	—	—	—
D	16	1	04	96	34	15
E	13	1	83	77	1B	D3
F	14	0	—	—	—	—

(c) Cache: Sixteen sets, 4-byte blocks, direct mapped

Select one:

1. VPN = 0x14, VPO= 0x0F, TLBI= 0x3, TLBT= x03, TLB Hit? =Y, Page Fault?= N, PPN = 0x0D, PTAR: N
2. VPN = 0x0F, VPO= 0x14, TLBI= 0x6, TLBT= x06, TLB Hit? =N, Page Fault?= N, PPN = 0x0F, PTAR: N
3. VPN = 0x0F, VPO= 0x14, TLBI= 0x3, TLBT= x03, TLB Hit? =Y, Page Fault?= N, PPN = 0x0D, PTAR: N
4. VPN = 0x0F, VPO= 0x14, TLBI= 0x6, TLBT= x03, TLB Hit? =N, Page Fault?= N, PPN = 0x0D, PTAR: N

The correct answer is: VPN = 0x0F, VPO= 0x14, TLBI= 0x3, TLBT= x03, TLB Hit? =Y, Page Fault?= N, PPN = 0x0D, PTAR: N

Question 275
Not answered
Marked out of 5.00

Identify a high level language the uses explicit allocator ....

Select one:

- 1. Lisp
- 2. Java
- 3. C
- 4. ML

The correct answer is: C

Question 276
Not answered
Marked out of 10.00

Select two mechanisms for changing control flow ....

Select one or more:

- 1. push
- 2. jmp
- 3. call
- 4. pop

The correct answers are: jmp, call

Question 277
Not answered
Marked out of 10.00

The task of converting a virtual address to a physical one is known as ....

Answer:

The correct answer is: address translation

Question 278
Not answered
Marked out of 25.00

Consider the following program containing Non-local jumps and Signal handler.

Suppose that the user presses a key combination of Ctrl & C after 700 secs.

Write the output produced by the program in the first 1000 secs.

```
jmp_buf buf;

void handler(int sig)
{
   longjmp(buf, 1);
}
int main()
{
   if (!setjmp(buf)) {
      Signal(SIGINT, handler);
            puts("STARTING LECTURE VIDEO\n");
   }
   else
         puts("YES - UNDERSTANDING ....!!!!!!!\n");
   while(1) {
            sleep(300);
            puts("YES - WATCHING &
LISTENING????\n");
   }
   exit(0);
}
```

Select one:

1. STARTING LECTURE VIDEO
YES - WATCHING & LISTENING????
YES - WATCHING & LISTENING????
YES - WATCHING & LISTENING????

2. STARTING LECTURE VIDEO
YES - WATCHING & LISTENING????
YES - UNDERSTANDING ....!!!!!!!
YES - WATCHING & LISTENING????
YES - WATCHING & LISTENING????

3. STARTING LECTURE VIDEO
YES - WATCHING & LISTENING????
YES - WATCHING & LISTENING????
YES - UNDERSTANDING ....!!!!!!!
YES - WATCHING & LISTENING????

4. STARTING LECTURE VIDEO
YES - WATCHING & LISTENING????
YES - WATCHING & LISTENING????
YES - WATCHING & LISTENING????
YES - UNDERSTANDING ....!!!!!!!

The correct answer is: STARTING LECTURE VIDEO
YES - WATCHING & LISTENING????
YES - WATCHING & LISTENING????
YES - UNDERSTANDING ....!!!!!!!
YES - WATCHING & LISTENING????

Question 279

Not answered

Marked out of 5.00

The ID number of x86-64 System call write is ………

Select one:

1. 1
2. 0
3. 3
4. 2

The correct answer is: 1

Question 280

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and Identify atleast one feasible output

```
int main()
{
    printf("L0\n");
    fork();
    printf("L1\n");
    fork();
    printf("Bye\n");
}
```

Select one:

1. L0
Bye
L1
Bye
L1
Bye
Bye

2. L0
L1
L1
Bye
Bye
Bye
Bye

3. L0
Bye
Bye
L1
Bye
L1
Bye

4. L0
Bye
Bye
L1
Bye
Bye
L1

The correct answer is: L0
L1
L1
Bye
Bye
Bye
Bye

Question 281
Not answered
Marked out of 5.00

Identify Class of thread-unsafe functions: ………………..

Select one:
1. Functions that do not call thread-unsafe functions
2. Functions that do not return a pointer to a static variable
3. Functions that do not keep state across multiple invocations
4. Functions that do not protect shared variables

The correct answer is: Functions that do not protect shared variables

Question 282
Not answered
Marked out of 10.00

Say TRUE or FALSE

(i) Signal Handler can be interrupted by other signal handlers ……………………

(ii) Signal Handlers cannot be nested ……………………………….

Select one:
1. FALSE, FALSE
2. TRUE, FALSE
3. TRUE, TRUE
4. FALSE, TRUE

The correct answer is: TRUE, FALSE

Question 283
Not answered
Marked out of 5.00

………………… function is used by Linux process to create new areas of virtual memory and to map objects into these areas.

Select one:
1. vmap
2. pmap
3. munmap
4. mmap

The correct answer is: mmap

Question 284

Not answered

Marked out of 20.00

Consider Virtual Address to Physical address translation on a small system with a TLB and L1 D-cache with the following assumptions:

The memory is byte addressable.

Virtual addresses(VA) are 32 bits wide.

Physical addresses(PA) are 32 bits wide.

The page size is 4KB.

The TLB is four-way set associative with 1024 total entries.

The L1 D-cache is physically addressed and four-way associative mapped, with a 32-byte line(block)size and 4K total blocks.

Cache block is byte addressable.

Determine the number of bits in the following

VPN = ... bits, VPO = ... bits, TLBI = ... bits, TLBT = ... bits

Total Number of entries in Page Table, PTEs = ... entries

Select one:

1. VPN=20 bits, VPO=12 bits, TLBI=12 bits, TLBT=8 bits, PTEs= 1M entries

2. VPN=20 bits, VPO=12 bits, TLBI=8 bits, TLBT=12 bits, PTEs= 2M entries

3. VPN=12 bits, VPO=20 bits, TLBI=8 bits, TLBT=12 bits, PTEs= 1M entries

4. VPN=20 bits, VPO=12 bits, TLBI=8 bits, TLBT=12 bits, PTEs= 1M entries

The correct answer is: VPN=20 bits, VPO=12 bits, TLBI=8 bits, TLBT=12 bits, PTEs= 1M entries

Question 285

Not answered

Marked out of 30.00

Consider the following program wrongcnt.c showing improper synchronization between two threads (thread1, thread2) when accessing the global variable cnt in their corresponding thread function.

Show atleast SIX Safe Trajectories in the progress graph which lead to proper thread synchronization.

```
/* Global shared variable */
volatile long cnt = 0; /* Counter */

int main(int argc, char **argv)
{
   long niters;
   pthread_t tid1, tid2;

   niters = atoi(argv[1]);
   pthread_create(&tid1, NULL, thread, &niters);
   pthread_create(&tid2, NULL, thread, &niters);
   pthread_join(tid1, NULL);
   pthread_join(tid2, NULL);

   /* Check result */
   if (cnt != (2 * niters))
      printf("BOOM! cnt=%ld\n", cnt);
   else
      printf("OK cnt=%ld\n", cnt);
   exit(0);
}
/* Thread routine */
void *thread(void *vargp)
{
   long i, niters =   *((long *)vargp);

   for (i = 0; i < niters; i++)
      cnt++;

   return NULL;
}
```

Select one:

1. H1,L1,U1,S1,T1,H2,L2,U2,S2,T2
H1,H2,L1,U1,L2,S1,U2,S2,T2,T1
H2,L2,U2,H1,L1,U1,S2,S1,T1,T2
H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,L2,U1,U2,S1,S2,T2,T1
H1,L1,H2,L2,U1,U2,S1,S2,T1,T2

2. H1,L1,U1,S1,H2,L2,U2,S2,T1,T2
H1,H2,L1,U1,L2,S1,U2,S2,T2,T1
H2,L2,U2,H1,L1,U1,S2,S1,T1,T2
H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,U1,S1,L2,U2,S2,T2,T1
H1,L1,U1,S1,T1,H2,L2,U2,S2,T2

3. H1,L1,U1,S1,T1,H2,L2,U2,S2,T2
H1,H2,L1,U1,L2,S1,U2,S2,T2,T1
H2,L2,U2,H1,L1,U1,S2,S1,T1,T2
H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,L2,U1,U2,S1,S2,T2,T1
H1,L1,U1,H2,L2,U2,S1,S2,T1,T2

4. H1,L1,U1,S1,H2,L2,U2,S2,T1,T2
H1,H2,L1,U1,S1,L2,U2,S2,T2,T1
H2,L2,U2,S2,H1,L1,U1,S1,T1,T2
H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,U1,S1,L2,U2,S2,T2,T1
H1,L1,U1,S1,T1,H2,L2,U2,S2,T2

The correct answer is: H1,L1,U1,S1,H2,L2,U2,S2,T1,T2
H1,H2,L1,U1,S1,L2,U2,S2,T2,T1
H2,L2,U2,S2,H1,L1,U1,S1,T1,T2

H1,H2,L2,U2,S2,L1,U1,S1,T2,T1
H2,H1,L1,U1,S1,L2,U2,S2,T2,T1
H1,L1,U1,S1,T1,H2,L2,U2,S2,T2

Question 286

Not answered

Marked out of 10.00

A .... is a global variable with a nonnegative integer value that can only be manipulated by two special operations, called P and V:

Answer:

The correct answer is: semaphore

Question 287

Not answered

Marked out of 20.00

Given the following program, Draw the process graph and and Identify atleast one infeasible output.

```
int main()
{
 printf("L0\n");
   if (fork() != 0) {
       printf("L1\n");
       if (fork() != 0) {
           printf("L2\n");
               }
   }
   printf("Bye\n");
}
```

Select one:

1. L0, L1, Bye, Bye, L2, Bye
2. L0, L1, Bye, L2, Bye, Bye
3. L0, Bye, L1, Bye, Bye, L2
4. L0, Bye, Bye, L1, L2, Bye

The correct answer is: L0, Bye, L1, Bye, Bye, L2

Question 288

Not answered

Marked out of 5.00

In Producer-Consumer Problem implementation, semaphore .... enforces mutually exclusive access to the the buffer.

Inserting an item into a shared buffer: **PRODUCER THREAD FUNCTION**

```
/* Insert item onto the rear of shared buffer sp */
void sbuf_insert(sbuf_t *sp, int item)
{
    P(&sp->slots);                          /* Wait for available slot */
    P(&sp->mutex);                          /* Lock the buffer */
    sp->buf[(++sp->rear)%(sp->n)] = item;  /* Insert the item */
    V(&sp->mutex);                          /* Unlock the buffer */
    V(&sp->items);                          /* Announce available item */
}
```

sbuf.c

Removing an item from a shared buffer: **CONSUMER THREAD FUNCTION**

```
/* Remove and return the first item from buffer sp */
int sbuf_remove(sbuf_t *sp)
{
    int item;
    P(&sp->items);                          /* Wait for available item */
    P(&sp->mutex);                          /* Lock the buffer */
    item = sp->buf[(++sp->front)%(sp->n)]; /* Remove the item */
    V(&sp->mutex);                          /* Unlock the buffer */
    V(&sp->slots);                          /* Announce available slot */
    return item;
}
```

Select one:

1. sbuf
2. mutex
3. items
4. slots

The correct answer is: mutex

Question 289

Not answered

Marked out of 5.00

An .... is a transfer of control to the OS kernel in response to some event (i.e., change in processor state)

Answer:

The correct answer is: exception

Question 290

Not answered

Marked out of 10.00

The .... operation in V also occurs indivisibly, in that it loads, increments, and stores the semaphore without interruption.

Answer:

The correct answer is: increment

Question 291
Not answered
Marked out of 10.00

Function fork is ·····································.

Select one:
- 1. called once, never returns
- 2. called once, returns one or more times
- 3. called once, returns only once
- 4. called once, returns twice

The correct answer is: called once, returns twice